A Prescriptive Design for Trustworthy Online Identifiers: The bMail ID Approach

Abstract

Email addresses are widely used as personal online identifiers (IDs) on e-commerce and e-services platforms owing to their global reach and communicability. However, the current email-based ID infrastructure suffers from inherent design limitations stemming from the lack of sender identifiability and verifiability, which undermines digital trust. In addition, the expiration or termination of email addresses can result in incompatibility with registered online member IDs, whereas the rigid ID management practices on e-commerce platforms often prohibit updating user IDs. To overcome these challenges, we propose the Bright eMail (bMail)-based ID system, a novel architecture that enhances the sender-side individual trust and the receiver-side social trust by providing the properties of identifiable anonymity, verifiability, sustainability, and changeability. A cross-national survey of user opinions across South Korea, the United States, and China demonstrates that the proposed features of the bMail ID system are perceived as necessary across countries. We further examine the role of culture and alternative telecommunication systems by analyzing cross-country variations. Based on these findings, we design a prescriptive architecture for the bMail ID system that facilitates both trustworthy email services and robust member IDs for online platforms. This research contributes to design science by proposing a critical infrastructure that supports the establishment of preventive cybersecurity grounded in the Bright Origin Principle.

Keywords: Email Identifier; Online Identifier; Bright eMail; bMail ID; bMember; Identifiable Anonymity

1. INTRODUCTION

Email is one of the primary telecommunication channels in our daily lives and business. Recently, email addresses have been widely adopted as *Online Identifiers* (commonly abbreviated as Online IDs) on e-commerce and e-service platforms (Mueller et al., 2006). Email is convenient for both telecommunication and ID purposes, but it faces major challenges from a cybersecurity perspective. From a receiver's point of view, received email IDs may be fabricated; some emails may be phishing attempts; and malicious emails may cause hacking, data breaches, and destruction of systems. On the other hand, from the sender's point of view, legitimate emails may not be fully trusted and may require the sender to bear the burden of proving their trustworthiness (Roghanizad & Bohns, 2017). Given the benefits and risks associated with email-based ID systems, there is a critical need for an alternative architecture that ensures trustworthiness for senders and safety for receivers. To this end, we need to identify the necessary features for alternative ID systems, in addition to the generic beneficial features of email-based IDs, such as *communicability, global reach, memorability, optional anonymity,* and *optional recognizability*.

To address this issue, we propose the Bright eMail-based ID system (hereafter, the bMail ID system), which integrates the Principle of Identifiable Anonymity from the principles of the Bright Internet (Lee, 2015) with the properties of email-based ID systems. *Identifiable anonymity* implies that senders voluntarily agree to be identified to gain trust while they may retain anonymity to protect their privacy (Lee et al., 2018). Thus, a sender with a bMail-based ID can be identifiable and trustworthy to unknown email receivers, without sacrificing the sender's privacy.

Another challenge with email IDs is the need for verification to prevent malicious attempts such as phishing and malware attacks. Generative AI and deepfake technologies further exacerbate these risks (Vecchietti et al., 2025). However, the current email system does not have the capability to verify whether the sender is genuine or fraudulent (Shen et al., 2021). Because fake senders may mislead recipients into fraudulent verification channels, a backup channel to verify the sender's authenticity is essential for avoiding sophisticated phishing attempts. Therefore, *verifiability* is a critical feature that bMail-based IDs should be equipped with.

The third challenge arises from the discontinuation of an email-based ID when the email service is terminated or no longer accessible. For instance, students graduate from universities and employees change companies; consequently, their email addresses affiliated with these institutions are no longer available (University of Minnesota, 2024). Although this issue may be avoided by using email addresses from third-party email services, it cannot be fully resolved, as such services may discontinue their operations for various reasons. Therefore, online IDs should be *sustainable,* remaining usable as long as users wish to continue using them.

Finally, e-commerce platforms that adopt email addresses as IDs need to provide the flexibility to update email-based IDs, not only for users' convenience but also to retain their customers. Because most platforms do not permit changing member IDs to avoid fraudulent manipulation, this creates a serious challenge when a registered email ID used across multiple platforms is terminated. Thus, e-commerce platforms need to establish a secure ID change protocol with email ID providers, which is not available in existing email services. Therefore, the bMail ID system needs to incorporate *changeability* so that a changed email ID can be updated through a secure collaborative protocol between the bMail ID providers and e-commerce platforms.

This research sets out to conceptualize, validate, and design the features required for the bMail ID system to ensure the following four properties.

(1) *Identifiability*: Ensure the identifiability of the sender's primary ID, even if anonymity of the sender's online IDs is optionally allowed.

(2) *Verifiability*: Provide a cross-referencing communication channel through the bMail service provider, allowing receivers to verify the sender's authenticity even if the receiver does not personally know the sender's verification address. This is especially necessary when the sender's email ID is compromised and used for phishing attempts.

(3) *Sustainability*: Guarantee a lifelong email ID so that a changed email address does not force a change in the member IDs already registered across multiple e-commerce platforms.

(4) *Changeability*: Allow the flexibility to update user IDs on e-commerce platforms through a secure protocol in collaboration with the bMail ID service provider if the registered email ID is no longer operational.

In the Information Systems literature, it has long been reported that users do not always adopt technologies merely because of their perceived ease of use and usefulness (Venkatesh et al., 2003; Kim & Kankanhalli, 2009). Thus, we examine whether the aforementioned prescriptive features incorporated into the bMail ID system, along with the generic properties of email-based ID systems, influence people's adoption intention. To this end, we theorize that the design goals, derived from both sender and receiver perspectives to gain individual trust and build social trust, mediate the effects of the bMail ID properties on user adoption as illustrated in Figure 1. This research model provides user-centric empirical support for the design of the proposed bMail ID system by validating the importance of these key features from potential users' perspectives.

To validate the research model for bMail ID adoption, we conduct a cross-national survey of user opinions from three countries with varying internet cultures and institutions: South Korea, the United States, and China. Based on the survey data, we examine the importance of the prescriptive features in the design of bMail-based ID infrastructure. In addition, we validate the necessity of these features by identifying their relevance to real-world cybersecurity threats through text mining of reported cybersecurity incidents in the United States. To formally test the proposed research model of bMail ID adoption, we employ Structural Equation Modeling (SEM). Furthermore, we discuss the role of national cultures and alternative telecommunication systems in predicting the preference for adopting the bMail ID system by analyzing cross-country variations.

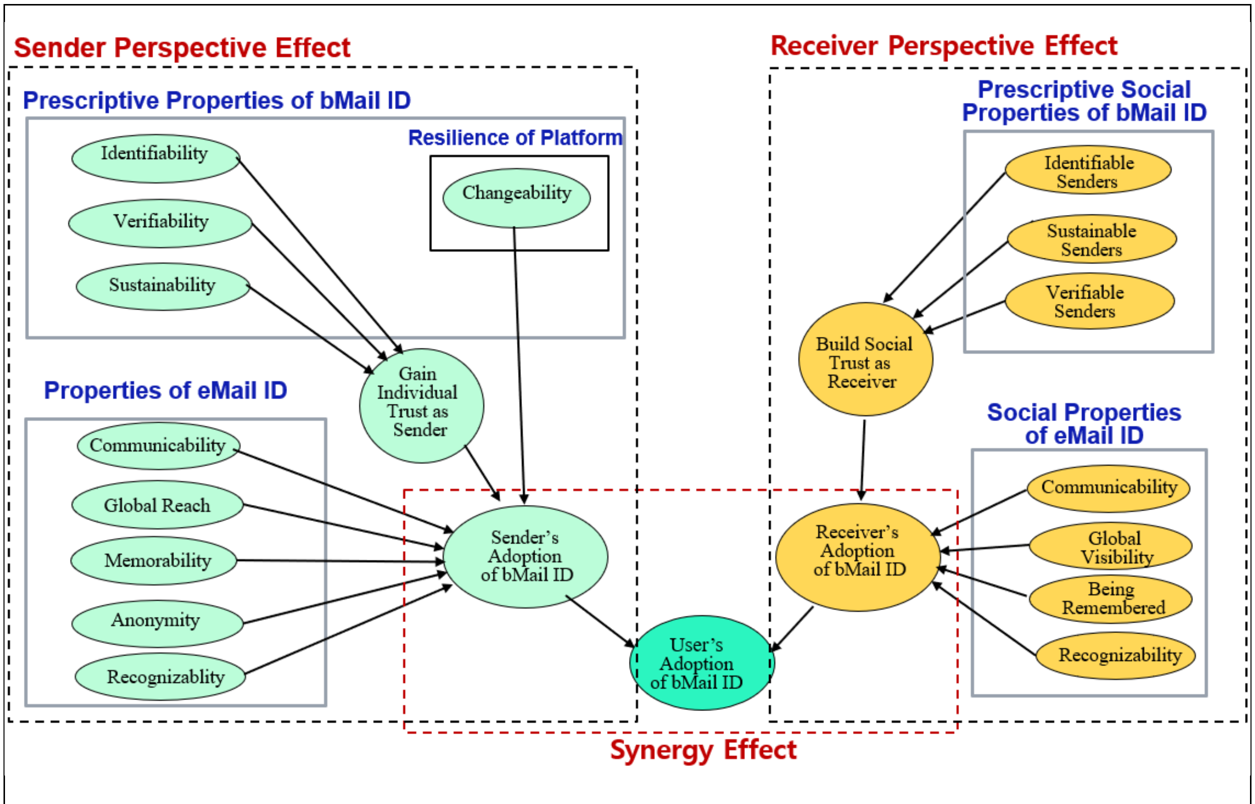

Figure 1. Research Model for bMail ID Adoption

This study follows the design science research guidelines (Hevner et al., 2004) as summarized in the conclusion (Table 9). To this end, the remainder of this paper is organized as follows. Section 2 reviews fake online IDs, evidence of ID-related threat types, and characterizes the categories of digital IDs. In addition, we distinguish the concepts of identification, authentication, identifiability, and accountability, and contrast them with the notion of Identifiable Anonymity adopted in the bMail ID system. A text mining analysis is conducted using a cybersecurity incidents database. Section 3 derives five generic properties of conventional email IDs and proposes four prescriptive properties for bMail IDs from the sender and receiver perspectives. Section 4 proposes a research model that will be validated through Structural Equation Modeling based on the surveyed opinions of business-experienced netizens. The constructs and variables are identified from the perspectives of both senders and receivers, and eight hypotheses are then proposed to validate the identified properties. Section 5 presents survey results that capture the descriptive evaluation of the email ID system and prescriptive evaluation of the proposed bMail ID system from respondents in three countries. Section 6 analyzes the structural equation models and tests the hypotheses from the sender and receiver perspectives, demonstrating the robustness of the results across the three countries. Country effects are discussed with respect to cultural differences and the penetration of alternative telecommunication channels. Section 7 designs the architecture of the bMail-based ID infrastructure by identifying the functions of key stakeholders. Finally, Section 8 summarizes the findings according to the design science research guidelines and discusses their theoretical and practical implications.

2. LITERATURE REVIEW AND FEATURES OF THE bMail ID SYSTEM

Most e-tailers (such as Amazon in the USA, Alibaba in China, and Coupang in South Korea), portals (such as Google in the USA, Baidu in China, and Naver in South Korea), and social networks and messaging services (such as Facebook in the USA and KakaoTalk in South Korea) adopt users' email addresses as online member IDs. Some of them allow phone numbers as alternative IDs. An exception is WeChat in China, which heavily depends on QR codes along with phone numbers. The proliferation of these email IDs demonstrates the necessity of the bMail ID infrastructure. Let us review the seriousness of fake online ID issues and discuss why the principle of identifiable anonymity and the features of the bMail-based ID system are necessary.

2.1 Fake Online Identities

As digital transactions surpass conventional ones, online identity fraud cases account for more than 70% of all identity fraud incidents in 2023 (Javelin Strategy & Research, 2024). Various fraud types, such as identity theft, identity spoofing, impersonation attacks, and account takeovers, entail the misuse of others' identities. Fake IDs or fake accounts may exploit user profiles to register for online platform services to generate fake reviews and misinformation (He et al., 2022) and to conduct cyberbullying to harm other online users (Lareki et al., 2023). Consumers' losses from identity fraud incidents totaled $27.2 billion in 2024, according to the Javelin Identity Fraud Study (Pitt, 2025).

Current identity fraud defenses rely primarily on authentication at account creation, login, and continuous behavioral verification; however, these methods have become less reliable amid large-scale data breaches and increasingly sophisticated fraud tactics (Kim et al., 2026). Users' behaviors are monitored to detect identity fraud in loan transactions (Xu et al., 2022) and in online reviews (Wu , Z. et al., 2024). Although these approaches can improve fraud detection, they are inherently reactive because they depend on signals that emerge only after accounts have already been created and used to cause harm.

Recognizing these limitations, we advocate a proactive defense paradigm that focuses specifically on preventing identity fraud. Given that email was used as the contact method in 25% of fraud cases (Federal Trade Commission, 2024), we propose the *Bright eMail* (denoted as *bMail*) ID approach as a preventive solution in our study.

2.2 Evidence of ID-Related Threat Types and Corresponding Prescriptions

We study the empirical evidence and cybersecurity incidents regarding the features of the bMail ID system as summarized in Table 1. Based on these observations, we identify the need for four prescriptive design requirements as the key features of the proposed bMail ID system. The empirical evidence described below is presented in the fourth column of Table 1.

(1) Need for Identifiability: Amazon remains the top platform for fake reviews, which highlights the need for identifiability to reduce such fraudulent activities (Paget, 2024).

(2) Need for Verifiability: Phishing emails are identified as a leading initial attack vector in harmful cyber-attacks, demonstrating the need for email receivers to verify suspicious senders whose accounts may be compromised (The Cybersecurity and Infrastructure Security Agency, 2022).

(3) Need for Sustainability: Email users at the University of Minnesota (2024) experienced serious discontinuity of their email IDs when students graduated and employees left the university. This kind of discontinuity is likely to occur across universities and institutions. To maintain the continuity of registered member IDs on e-commerce platforms, users need support for sustainable email IDs.

(4) Need for Changeability: Major platforms such as Naver and KakaoTalk do not allow users to change their user IDs even when their associated email accounts become unavailable (Inquivix, 2023; Naver, 2025). As a result, these platforms may lose communication with customers without recognizing that they are no longer reachable. Therefore, these platforms should allow legitimate users to change their IDs through a secure and authorized process.

Table 1. ID-Related Threat Types and Prescriptions

Prescriptions		ID-Related Threat Types		
bMail-based Initiative	**Online Platform-based Initiative**	**Type of Threat**	**Empirical Evidences**	**Incident Evidences in PRC**
Need for Identifiability	Recognize bMembers as identifiable customers	**Fake IDs:** **(Sender)** Email IDs are falsely compromised; (**Receiver**) Fake IDs are used for false reviews.	Fake reviews at Amazon (Paget, 2024).	25,928 incidents of ID thefts and fake IDs
Need for	Recognize	**Phishing Attempts:**	Phishing attacks	2,104 incidents

Verifiability	bMembers as verifiable customers	(**Sender**) Email IDs are abused for phishing attempts; (**Receiver**) Fake IDs deceive email receivers.	(CISA, 2023).	of phishing attempts
Need for Sustainability	Retain lifelong bMember IDs	**Discontinued email account:** Email IDs are not sustainable.	Universities discontinue their email IDs (UMN, 2024).	Not Applicable
Need for Changeability	Support the changeability of discontinued IDs	**Need to change expired email IDs:** Platforms should allow legitimate changes of IDs.	Platforms do not allow users to change IDs (Inquivix, 2023; Naver, 2025).	Not Applicable

Text Mining Analysis. We also conducted a text mining analysis to derive incident-based evidence for the period from 2005 to 2025 in the Privacy Rights Clearinghouse (PRC) database to identify the necessary features, as shown in the fifth column of Table 1. The details of the text-mining process are described in Appendix A. The keywords used to detect identifiability and verifiability are shown in Appendix Table A1. The keywords for identifiability capture incidents in which senders' true identities were concealed, stolen, or unidentifiable, including terms related to unauthorized access, impersonation, compromised credentials, and the physical theft of devices containing personal records. The keywords for verifiability capture two distinct failure modes: deceptive origin incidents where malicious senders misled receivers through phishing, spoofing, or fraudulent communications; and inadvertent disclosure incidents where content reached unintended receivers due to system errors or human mistakes. Both reflect a common limitation in which the receivers are unable to confirm whether the communication and its source are legitimate.

According to Latent Dirichlet Allocation (LDA) topic modeling applied to 81,847 incidents, 28,032 were classified as relevant to bMail feature requirements, including 25,928 (92.49%) related to identity theft and fake IDs, and 2,104 (7.51%) related to email phishing. These findings support the need for identifiability and verifiability (Appendix Table A2). Since sustainability and changeability do not directly fall under the cybersecurity incident category, they are not detected from the PRC database.

2.3 Identification, Authentication, Identifiability, and Accountability

To conceptualize our research objective, we need to distinguish the notions of *Identification, Authentication, Identifiability*, and *Accountability*. For most systems, identification and authentication serve as the first line of defense. Identification is the process by which a user claims an identity, essentially stating, "This is who I am" (Kim et al., 2026). Authentication, on the other hand, verifies the legitimacy of the claimed identity, confirming, "You are who you claim to be." Therefore, authentication alone is not sufficient to verify whether the claimed identity is genuine. Thus, we need to incorporate identifiability in our study to allow receivers to legitimately verify the sender's genuine identity even though the email name may be expressed as a pseudonym.

Accountability means the ability to trace actions back to a responsible individual or entity, making it possible to seek compensation, particularly in legal or financial contexts (Feigenbaum et al., 2014). Therefore, accountability requires additional financial data and credit records that are not possible to trace in the context of email exchanges. In this regard, identifiability is a foundational step necessary to ensure accountability, and we limit the target of this research to identifiability.

In electronic markets, fake email IDs are often abused to generate fraudulent customer reviews on retailer sites. For example, Amazon reported that up to 42% of customer reviews might be fake (Shukla & Goh, 2024). Fake IDs are also responsible for generating false news that can harm the reputations of individuals and companies, and may even influence global politics. Although online platforms make efforts to delete fake reviews and news, the damage to consumers and society has often remained (He et al., 2022). Therefore, our study aims to enable the identifiability of reviewers, while allowing anonymity to protect their privacy under the notion of identifiable anonymity.

2.4 Identifiable Anonymity Approach in the bMail ID System

The pros and cons of anonymity and non-anonymity have long been debated. It is reported that social media users exhibit more uncivil behavior in anonymous settings than in non-anonymous environments (Qian et al., 2025). Anonymity also affects users' perceptions of message credibility, and anonymous messages are regarded as less trustworthy (Chesney & Su, 2010).

Conversely, visible identity information can increase consumer trust in reviews and influence purchasing behavior (Forman et al., 2008). However, disclosed identities in online communities may discourage user participation, particularly among individuals who wish to critique persons and products confidentially. Identity verification protocols may deter users from writing reviews due to privacy concern (Shukla and Goh, 2024). Shifting from anonymity to identity disclosure inhibits users' willingness to contribute content, although this shift generates higher quality content (Pu et al., 2020).

It is interesting to note that social platforms Meta and Twitter (X) offer paid verification services that allow users to signal profile authenticity and build credibility with their audiences (Meta, 2025; X Corp, 2025). Meta requires users to use real names, while X allows users to use pseudonyms. To harmonize this conflict between identifiability and anonymity, we adopt the Principle of Identifiable Anonymity (Lee, 2015; Lee et al., 2018) in the bMail ID system. These two paid services demonstrate the potential business value of bMail ID services that offer the option of identifiable anonymity.

To implement the notion of identifiable anonymity in the context of the bMail ID system, email names may allow users to use pseudonyms (such as 19905217@gmail.com) to hide their real names. However, the domain names of bMail can assure bMail identifiability if the email service providers extend their services with bMail-based domain names such as bGmail from Gmail, bCoremail from Coremail, and bNaver from Naver. Since most email service providers already adopt authentication protocols such as DKIM (Domain Keys Identified Mail), SPF (Sender Policy Framework), and similar protocols, the domain names of bMail themselves cannot be illegally spoofed at the transport layer of email transmission.

Thus, if the email sender wishes to establish trust with receivers, the sender may voluntarily guarantee their identifiability by adopting a bMail account that is certified by a trusted third party. The trusted third parties, such as telecommunication service providers or banks, naturally maintain users' genuine identities by the nature of their business. Therefore, the bMail ID system balances identifiability to ensure trustworthiness for email receivers, while anonymity may be optionally selected if the sender does not want to disclose private information. It is the sender's choice whether to use an anonymous email name (such as 19905217@bgmail.com) or a real name (such as efraim.turban.hawaii@bgmail.com).

2.5 Categories of Digital IDs

To understand the locus of email-based IDs in comparison with other types of digital IDs, we categorize digital ID systems into three layers.

(1) **Governmental IDs**. The primary governmental ID is the national ID, which identifies an individual's name, birth date, address, and face as identifiers. South Korea and China have national resident identity card systems. In this case, the driver's license and passport are secondary governmental IDs. However, the USA uses the driver's licenses, Social Security Numbers, and passports as primary decentralized governmental IDs. Governmental IDs contain highly sensitive personal information, and the illegal disclosure of such information poses significant risks to users. In this sense, governmental IDs should be used only for governmental services and are not suitable as public online IDs.

(2) **Institutional IDs**. Institutional IDs are issued by trusted institutions such as mobile phone companies, banks, hospitals, corporations, and universities. The issuance of these IDs is usually authenticated by referencing governmental IDs and verifying the applicants in person. These IDs may be used for the institution's own services, but they may have a limited service period and do not guarantee sustainability as public online IDs.

(3) **Tertiary public online IDs**. Tertiary online IDs include email addresses, phone numbers, and QR codes that are issued by service providers. Unlike governmental or institutional IDs, tertiary online IDs are user-initiated for online use and do not necessarily require reference to governmental or institutional IDs. The email-based IDs that we study fall into this category. However, the bMail-based ID is a tertiary ID that includes certification from trusted institutions such as telecommunication companies or banks.

Centralized vs. Decentralized ID Management Systems. Online identity models can be classified into centralized and decentralized ID management perspectives. Centralized ID models store user information on the service provider side, while decentralized ID models keep data on the user side.

Centralized ID models may adopt a federated ID model. A federated ID model has a single sign-on capability like Microsoft Passport (Microsoft, 1999; Liu et al., 2021) allowing users to access multiple sites with one identity. The bMail ID, as well as the email ID, may operate like a federated ID model depending on the user's choice. Users may simply use one ID across all platforms, or may use different IDs for different platforms to reduce the risk of concentration. Centralized IDs require platforms to store sensitive personal data, which may pose significant risks, as attackers may compromise centralized repositories to steal sensitive information in bulk (Sen & Borle, 2015).

The bMail ID system can achieve the effect of a federated ID system, but the registration of IDs on e-commerce platforms is not centralized as it is in a federated ID system and the private data stored in the certifying institutions are not copied to the servers of the bMail service provider. The architecture of the bMail ID infrastructure will be presented in Figure 3 in Section 7. In this sense, the bMail ID system is not as risky as a traditional centralized ID system.

Decentralized ID models may adopt either a user-centric ID or Self-Sovereign Identity (SSI) (Allen, 2016). A user-centric ID model refers to providing users with control over their identity providers and enabling them to share attributes without requiring verified trust between platform service providers and identity providers. Login options provided through Facebook and Google are examples of this model. This scenario differs from the bMail service framework because banks and phone service providers already have users' personal data for their businesses, and the bMail service provider will only issue a certification of identifiability without storing the personal data on its own site to avoid potential risks of accidental disclosure.

On the other hand, the SSI model, supported by blockchain technology, is an emerging approach for ID evolution. However, in SSI, losing cryptographic keys can lead to an unrecoverable loss of data and funds due to the non-custodial feature of blockchain technology (Kshetri, 2017). Effective key management is critical for widespread adoption of SSI because the transition to an SSI infrastructure requires substantial changes rather than simple upgrades (Yildiz et al., 2021). The key mechanism for SSI does not support the capability of email-based IDs; thus, the bMail ID system aligns with the centralized ID model by using a separate certificate authority.

3. TWO DIMENSIONAL FEATURES OF THE bMail ID SYSTEM

We propose the integrated features of the bMail ID system by combining the generic properties of email-based IDs with the prescriptive properties of the bMail-based IDs. Second dimension is the sender and receiver perspectives. To this end, we distinguish interpersonal trust directed toward specific actors and institutional trust grounded in structural assurances (McKnight et al., 2002). In online environments, trust is directed not only toward interaction partners but also toward the broader institutional context that supports those interactions (Gefen et al., 2003). We pursue these two forms of trust in the context of identity-based systems from the perspectives of senders and receivers. Individual trust corresponds to interpersonal trust, as senders need to establish their credibility in direct interactions. Social trust corresponds to institutional trust, reflecting receivers' confidence in the overall communication environment and societal-level safeguards.

Based on this distinction, we conceptualize the bMail ID system as a two-dimensional framework capturing trust formation from both sender and receiver perspectives. These two dimensions are operationalized as individual trust and social trust, respectively, as contrasted in Table 2. Each cell will constitute a construct in Figure 1.

Table 2. Two Dimensions of Features

	Sender Perspective	**Receiver Perspective**
Generic Properties of eMail-based ID	Communicability with receivers Global reach to receivers Memorability of own IDs Optional anonymity to receivers Optional recognizability to receivers	Communicability with senders Global visibility to senders Being remembered by senders (Anonymous senders provide no value) Recognizable senders
Prescriptive Properties of bMail-based ID	Identifiability (gains individual trust) Verifiability (gains individual trust) Sustainability of the ID owner Changeability of eMember IDs	Identifiable senders (build social trust) Verifiable senders (build social trust) Sustainable senders Update eMember IDs at platforms

3.1 Sender Perspective Effect

3.1.1 Sender-Side Properties of eMail IDs

The construct, *sender-side properties of eMail IDs,* encompasses communicability, global reach, memorability, optional anonymity, and optional recognizability.

1) Communicability: Email was inherently designed as a communication tool. The State of the Workplace Communication reported that email remains the most popular communication tool, with 25% of remote workers selecting email as their most preferred method of communication (Hoory, 2023). Compared to national IDs and institutional IDs, email offers a unique feature, similar to phone numbers, in that it serves both as an identifier and a channel for communication (Mueller et al., 2006). Thus, the first key property of email is its communicability.

2) Global Reach: Global reach is a powerful capability of email by its nature. The swift transmission of email makes it a preferred medium for global communication, regardless of the location of recipients (Dürscheid & Frehner, 2013) without incurring additional costs. Phone services tend to restrict the use of phone numbers to domestic users, making it difficult for foreigners to register for their services. Messenger services are popular, but their user coverage is usually regional rather than global. These restrictions hinder their potential for globalization. Thus, we identify global reach as the second key advantage of email compared to other forms of identification.

3) Memorability: An email ID is easier for users to remember than other IDs. One challenge in online ID management is that users must remember multiple IDs and passwords across different platforms. Forgotten passwords may be updated easily through a verification process, but forgotten IDs are hard to trace. In this regard, email-based IDs reduce the effort to memorize them (Mueller et al., 2006). This is the third key advantage of using an email address as an online ID.

4) Anonymity: An anonymous email name may be optionally selected for the privacy protection of senders. Senders may want to disclose minimal personal information by using pseudonyms. Anonymity also plays a crucial role in protecting reviewers' privacy, shielding them from intimidation (Shukla & Goh, 2024). Moreover, anonymity fosters more honest critical feedback and encourages greater user engagement on online platforms (Pu et al., 2020). Protecting the anonymity of innocent individuals is particularly important in online environments to ensure their freedom of speech. This is why identifiable anonymity is adopted in the Bright Internet principles (Lee et al., 2018). Thus, optional anonymity is another property of using email addresses as identifiers, contributing to both privacy protection and user engagement. However, the benefit of this property may undermine recognizability.

5) Recognizability: Showing real names in email addresses makes the sender highly recognizable to receivers. Some users, such as university professors, prefer to use their real names in their email addresses because it enables them to convey professionalism, trust, and credibility, especially in formal work-related communications. It enhances the sender's transparency and authenticity, thereby increasing the likelihood that receivers will engage in email exchanges. In some cases, it is essential for the email address to be recognizable to others, ensuring that it is not mistaken for spam and encouraging receivers to view it as trustworthy (Markman & Scott, 2005). Thus, optional recognizability is another key property of using an email address as an online ID to build individual trust. The choice between anonymity and recognizability is a matter of user choice.

3.1.2 Sender-Side Prescriptive Properties of bMail IDs

The construct, *sender-side prescriptive properties of bMail IDs,* encompasses identifiability, verifiability, sustainability, and changeability.

6) Identifiability: Senders may voluntarily guarantee their own identifiability to gain individual trust from receivers. Innocent senders who assure identifiability can be regarded as a bright origin and can be trusted by others (Lee et al., 2018), distinguishing themselves from fraudulent actors. In the bMail ID scenario, using a bMail domain such as *bgmail.com* can demonstrate that senders from this bMail domain are identifiable. The user's identifiability can enhance trustworthiness in online communities (Qian et al., 2025; Forman et al., 2008). Since most email service providers already adopt a combination of authentication protocols such as DKIM, SPF, DMARC (Domain-based Message Authentication, Reporting & Conformance), and ARC (Authenticated Received Chain), the domain names of bMails can be transmitted without the risk of illegal spoofing during message transmission.

The domain names of the bMail service should be publicized so as to be verifiable from the receiver side of the bMails. To this end, the bMail ID service community should establish a portal site that can verify valid bMail domain names and present them in an easily recognizable manner. The bMail service providers at the receiver side may automatically confirm whether the sender's domain is a bMail domain. This can filter out the risk of sending bMails from fake bMail domains. For instance, if an email service provider such as *gmail.com* augments its service, establishing a bMail ID domain such as *bgmail.com*, the bMail is identifiable regardless of whether the personal name is efraim.turban.hawaii@bgmail.com or 19905217@bgmail.com. However, a display name such as "Efraim Turban" <12345@gmail.com> may be fake because *gmail.com* is not a valid bMail domain.

7) Verifiability: This feature provides a backup verification channel that is stored by the bMail service provider but not disclosed to receivers. The verification channel may be a backup email address, messenger ID, or phone number of the genuine sender, which the bMail system operator can use when a receiver requests verification while confronting a suspicious phishing attempt. Sometimes, the original email address itself might be compromised and used to send phishing emails. If the sender's backup channel can be used to verify the genuine sender for a requesting receiver, the phishing email can be promptly detected, and the real sender can react by changing the password and blocking the breach.

8) Sustainability: Guaranteeing a lifelong email ID aligns with the goal of sustainability particularly in the context of e-commerce (Day, 2024). For researchers, Persistent IDentifiers (PIDs), such as DOIs and ORCID iDs, provide unique keys for people, places, and things. These identifiers enable accurate information mapping across systems by facilitating search, discovery, recognition, and collaboration (Haak et al., 2012). Open and non-proprietary PIDs have proven valuable for promoting information sharing within and across systems. Like PIDs, a lifelong bMail ID ensures continuity, enabling individuals' contributions to be correctly identified, discoverable, and recognized across digital platforms, thus building reputation and increasing trustworthiness (Dellarocas, 2003). Thus, sustainability emerges as another essential feature of bMail, enhancing trust by offering users a reliable, persistent online identity. A bMail ID may substitute the role of an ORCID iD, offering the additional benefit of global communicability.

We propose that the above three properties contribute to the mediating goal, "Gain Individual Trust as Sender" from the receivers' point of view.

3.1.3 Resilience of the E-Commerce Platforms

9) Changeability: This implies allowing legitimate users to change their email-based IDs when the email address is no longer available. When students graduate, professors change universities, or employees switch companies, their institutional email addresses are often terminated by the institutions (Liu et al., 2021). Consequently, when users registered for e-commerce platforms with such email addresses, the platform companies lose their communication channels with these members. Signing in with a new email ID is usually not allowed to avoid duplicate IDs. If platforms do not support secure ID changes, users may experience frustration or disengagement, leading to customer churn and disruption in relationship marketing (Morgan & Hunt, 1994). Thus, we propose a changeable bMail ID system as a strategy to enhance the resilience of user identification, ensuring continuity even when current email IDs are deactivated.

In this regard, changeability is a crucial feature that enables both platform resilience and continued user membership. This process should be facilitated by the bMail service providers through collaboration on a secure ID-changing protocol.

3.2 Receiver Perspective Effect

From the receiver's perspective, the key advantages of email IDs include communicability with senders, global visibility to potential senders, ease of being remembered by senders, and recognizability of senders when they use real names. However, sender anonymity does not provide any direct value to receivers. In addition, the desirable features of bMail IDs for receivers are identifiability, verifiability, and sustainability, whereas changeability is not a direct benefit for receivers. However, as bMail usage becomes widely propagated, social trust through receiving bMails can be established. Therefore, examining the receiver-side benefits from a social norm perspective is necessary as follows.

3.2.1 Social Norms in bMail ID Adoption.

Social norms refer to collectively understood rules or standards that guide and constrain individual behavior within specific social contexts. They reflect shared perceptions of socially approved behavior, serving both descriptive functions (what most people do) and injunctive functions (what most people approve or disapprove of). In cybersecurity contexts such as combating fake news, social norms have been shown to function as effective socio-technical remedies (Gimpel et al., 2021).

In the context of bMail ID adoption, social norms theory offers a valuable lens for explaining why receiver-side adoption behavior is critical. The introduction and use of bMail IDs inherently entail social interactions between senders and receivers. Consequently, receiver perceptions and behaviors will be shaped by the visionary social value of adopting bMail in establishing social trust. If bMail usage is widely propagated, email recipients will receive more bMails and become safer accordingly. Therefore, receivers' decisions to adopt bMail IDs will be influenced not only by the individual benefit of gaining trust, but also by their ethical vision of building a safe cyberspace. In this regard, the receiver-side perspective requires the formation of an expected social norm of trust and safety.

3.2.2 Receiver-Side Properties of eMail IDs

The construct, *receiver-side properties of eMail IDs,* encompasses communicability, global reach, memorability, optional recognizability but excludes anonymity as follows.

1) Communicability with senders: By the nature of email, receivers can communicate with senders or e-commerce platforms whenever they need.

2) Global visibility to senders: Receivers can be globally visible to senders so that they can reach the receivers' email IDs wherever they may be located.

3) Being remembered by senders: Receivers' email IDs can be easily remembered by senders through their email records. In e-commerce, repeat interactions and personalization are critical to success, as remembering past interactions fosters trust, recognition, and positive engagement (Flacandji & Krey, 2020). Thus, being remembered serves as a foundation for building strong relationships with trading counterparts and registered e-commerce platforms.

4) Anonymous emails received: Sender anonymity does not create any value for receivers.

5) Recognizable senders: Receivers can recognize senders through the email addresses when real names are used. Compared with phone numbers, email senders are easily recognizable. For example, recognizable universities and company domain names, together with real names, enhance sender recognizability and trustworthiness.

3.2.3 Receiver-Side Prescriptive Properties of bMail IDs

The construct, *receiver-side prescriptive properties of bMail IDs,* encompasses identifiability, verifiability, sustainability but excludes changeability as follows.

6) Identifiable Senders: Identifiability of senders enables receivers to trust senders even when they are new. Similarly, the identifiability of reviewers on e-commerce platforms increases consumer trust in reviews. Recall that the social platforms Meta and Twitter (X) offer users paid verification services to gain credibility with their audiences. Therefore, we posit that receivers will be able to socially trust the senders and reviewers if they use bMail IDs.

7) Verifiable Senders: If the bMail service supports verification, receivers can verify whether the sender is genuine when they encounter ambiguous emails. When a received email address is suspicious of being compromised and used for phishing, receivers can request verification from the bMail service provider instead of being misled by disguised contacts controlled by accomplices. Thus, verifiable bMail IDs can help receivers to avoid the risk of phishing attempts, reducing the resulting damage.

8) Sustainable Senders: Senders with lifelong bMail IDs are more trustworthy to receivers because they provide a consistent and reliable way to verify a sender's identity over time, reducing fraud and uncertainty. The effect is similar to that of an ORCID iD, which increases credibility for readers, while Google Scholar cannot avoid confusion caused by identical names of authors.

9) Changeability of a sender's bMail ID can maintain contact between customers and e-commerce platforms, but it does not provide direct value to the receivers of bMails. Thus, this property is not included as a feature for building social trust. The goal of "Building Social Trust" is regarded as a mediating variable in the receiver-side model.

There are many other aspects that may influence the adoption of the bMail ID system, such as the factors discussed in TAM and subsequent research (Davis, 1989). However, this study focuses on the effect of the properties of email and bMail IDs from the perspectives of senders and receivers. Based on the features discussed above, we establish the research model for the bMail ID design, with two mediating variables: "Gain Individual Trust as Sender" and "Build Social Trust as Receiver". The synergistic outcome of these models is the adoption of bMail IDs using both sender and receiver perspectives, as shown in Figure 1.

4. RESEARCH MODEL FOR THE ADOPTION OF bMail ID DESIGN

The research model in this study aims to identify the necessary features of the bMail ID system that will achieve the goals of senders gaining individual trust and receivers building social trust. To this end, we adopt the generic properties of email IDs and the prescriptive properties of bMail IDs as latent constructs that determine the mediating goals, as depicted in the structural equation model (Figure 2). These mediating trusts ultimately motivate the adoption of the bMail ID system.

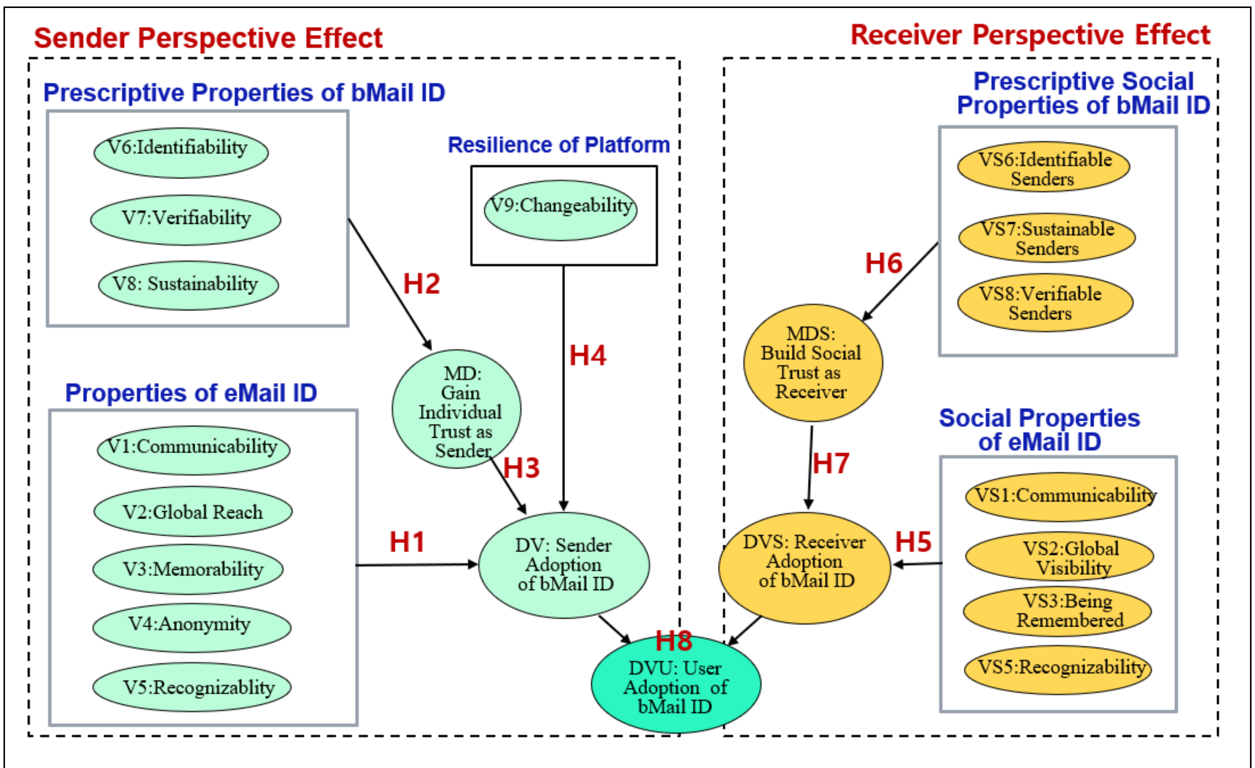

Figure 2. Structural Equation Model and Eight Hypotheses to Test

4.1 Constructs and Variables of the Structural Equation Model

The constructs and variables of this research model are defined in Table 3. From the sender's perspective, the latent construct 'Properties of email ID' is determined by five variables (V1, V2, V3, V4, V5). The latent construct 'Properties of bMail ID' is determined by three variables (V6, V7, V8), which motivate the mediating goal of 'Gain Individual Trust as Sender' (denoted as MD). The mediating goal, together with platform resilience through changeability (V9) and 'Properties of email ID', influences 'Sender Adoption of bMail ID' (denoted as DV) as a dependent variable.

Similarly, from the receiver's perspective, the latent construct 'Social Properties of email ID' is determined by four variables (VS1, VS2, VS3, VS5). The latent construct 'Social Properties of bMail ID' is determined by three variables (VS6, VS7, VS8), which motivate the mediating goal of 'Build Social Trust as Receiver' (denoted as MDS). The constructs 'Social Properties of email ID' together with 'Build Social Trust' influence the 'Receiver Adoption of bMail ID' (denoted as DVS) as a dependent variable. The adoptions as sender and receiver synergize to promote overall user adoption (denoted as DVU).

Table 3. Constructs and Variables of the Research Model

Perspectives	Constructs	Variables	
Sender Perspective	Properties of eMail ID	V1	Communicability
		V2	Global reach
		V3	Memorability
		V4	Anonymity
		V5	Recognizability
	Prescriptive Properties of bMail ID	V6	Identifiability
		V7	Verifiability
		V8	Sustainability
	Resilience of the Platform	V9	Changeability
	Gain Individual Trust as Sender (MD: Mediator of Individual Trust)	MD1	Trusted by email receivers
		MD2	Trusted by e-commerce counterparts
	Sender Adoption of bMail ID	DV	Sender Adoption
Receiver Perspective	Social Properties of eMail ID	VS1	Communicability with Senders
		VS2	Global Visibility to Senders
		VS3	Being Remembered by Senders
		VS5	Recognizable Senders

	Prescriptive Social Properties of bMail ID	VS6	Identifiable Senders
		VS7	Verifiable Senders
		VS8	Sustainable Senders
	Build Social Trust as Receiver (MDS: Mediator of Social Trust)	MDS1	Trust in email senders
		MDS2	Trust in e-commerce counterparts
	Receiver Adoption of bMail ID	DVS	Receiver Adoption
Synergy Perspectives	User Adoption of bMail ID	DVU	User Adoption

4.2 Propositions of Hypotheses

To test the validity of this research model, we propose eight hypotheses as marked in Figure 2. These hypotheses will be tested using the prescriptive opinions of netizens in South Korea, the United States, and China.

1) Effect of Sender-side Properties of Email ID

The construct, sender-side properties of email IDs, is represented by five variables (V1: communicability, V2: global reach, V3: memorability, V4: optional anonymity, and V5: optional recognizability). These properties contribute to the senders' adoption of bMail IDs. Thus, we propose Hypothesis H1.

H1: *The properties of email IDs positively contribute to the senders' adoption of bMail IDs.*

2) Effect of Sender-side Properties of bMail ID Requirement

The sender-side properties of bMail IDs are represented by three variables (V6: identifiability, V7: verifiability, V8: sustainability). They are necessary requirements for making the bMail IDs trustworthy. Thus, we posit that the mediating variable, 'Gain Individual Trust as Sender,' is achieved by these three bMail ID properties. Thus, we propose Hypothesis H2.

H2: *The properties of bMail IDs positively contribute to gaining individual trust as senders.*

The gained trust will eventually mediate the relationship between sender-side properties and the adoption of bMail IDs as senders, as proposed in Hypothesis H3.

H3: *Gaining individual trust as senders positively influences senders' adoption of bMail IDs.*

3) Effect of Resilience of Online Platforms

Users registered on e-commerce with an email ID may encounter a situation in which their registered email ID is suspended. To overcome this situation, we propose that online platforms offer a resilient option that allows members to securely change their IDs when the email ID is no longer in service. In this case, the bMail service provider needs to facilitate a secure change of the online platform's email IDs. The benefit of a changed ID is to retain platform customers while reducing unreachable members. Therefore, we propose Hypothesis H4.

H4: *The platform's changeability of email IDs, in collaboration with the bMail service provider, positively influences senders' adoption of bMail IDs.*

4) Effect of Receiver-side Social Properties of Email IDs

We examine the receiver's perspective, focusing on the social properties of receiving email IDs. The receiver-side properties of email IDs are measured by four variables (VS1: communicability with senders, VS2: global visibility to senders, VS3: being remembered by senders, and VS5: recognizable senders). We posit that these properties contribute to the social properties of bMail IDs. Thus, we propose Hypothesis H5.

H5: *The social properties of email IDs positively influence receivers' adoption of bMail IDs.*

5) Effect of Receiver-side Social Properties of bMail ID Requirement

The receiver-side social properties of bMail IDs are represented by three variables (VS6: identifiable senders, VS7: verifiable senders, and VS8: sustainable senders). As bMail senders become widely propagated, received bMails become socially trustworthy. Thus, we posit that the mediating goal, 'Build Social Trust as Receiver,' will be achieved by the propagation of bMail IDs. Thus, we propose Hypothesis H6.

H6: *The social properties of bMail IDs positively contribute to building social trust as receivers.*

The established social trust will mediate the relationship between receiver-side properties and the adoption of bMail IDs as receivers, as proposed in Hypothesis H7.

H7: *Establishing social trust as receivers positively influences receivers' adoption of bMail IDs.*

6) Synergistic Effects of Sender and Receiver Perspectives on Users' bMail ID Adoption

Based on the preceding hypotheses addressing both senders' and receivers' perspectives, we posit that users' adoption of bMail IDs is driven by the synergistic effects of senders' individual benefits and receivers' social benefits. Accordingly, we propose Hypothesis H8.

H8: *The sender-side adoption of bMail IDs and receiver-side adoption of bMail IDs positively influence users' synergistic adoption of bMail IDs.*

5. DATA SURVEY FROM THREE COUNTRIES

To test the hypotheses, we designed questionnaires as depicted in Appendix Table B1. All questionnaire items were rated on a seven-point Likert scale. To ensure the validity and reliability of the items, a pre-test with 80 respondents was conducted, and the questionnaires were refined based on their feedback. To collect robust cross-country data, we surveyed the opinions of netizens with work experience using professional online survey panels in three countries: SurveyMonkey in the USA, WJX (Wenjuanxing) in China, and Open Survey in South Korea. The validity of these survey platforms is supported by prior research (Thomas et al., 2025; Wu, T. et al., 2024; An et al., 2026). The survey was initially designed in English and subsequently translated into Korean and Chinese by bilingual co-authors for the regional surveys. The numbers of valid responses collected were 387 from South Korea, 440 from the USA, and 421 from China.

Appendix Table B2 summarizes the demographic information of survey respondents. Approximately 53% of respondents are female, while 47% are male. Overall, 67% of respondents reported engaging in international online activities, although this proportion varies across the three countries. The effects of these demographic characteristics are treated as control variables to enhance the external validity of our study (Luo et al., 2020).

6. SEM ANALYSIS AND TEST OF HYPOTHESES

To analyze the measurement quality and to test the research hypotheses, we adopt SmartPLS 4.0 software. SmartPLS is suitable for our study because it supports the study of formative constructs, does not require normal distribution assumptions, and handles interval-scale Likert scores (Hair et al., 2022). As the structural relationships are stable across countries according to our tests, pooling the data does not introduce systematic bias in hypothesis testing. The results of the SEM model using pooled data are summarized in Table 4, and country-level data in Table 5. We performed bootstrapping with 10,000 resamples to compute the *t*-statistics for all structural paths in the research model. The eight hypotheses tests based on these results indicate that the hypothesized path coefficients are statistically significant and directionally consistent across the three countries' data.

The significance of the variables for the bMail property constructs for each country is reported in Appendix Table C1. Multicollinearity is not a concern in this study as all Variance Inflation Factor (VIF) values are below the recommended threshold of 10 (MacKenzie et al., 2011; see Appendix Table C2). To address potential Common Method Variance (CMV), we adopted two diagnostic approaches: VIF assessment (Appendix Table C2) and the marker variable approach (Appendix Table C3). Both approaches indicate that CMV does not inflate the observed relationships among the constructs.

6.1 Tests of Hypotheses

Based on the collected data, we test each hypothesis using the pooled data in Table 4 and country-level data in Table 5. We found that all hypotheses are significantly supported in both the pooled data and the three country-level datasets as described below.

1) **Test of Hypothesis H1.** The construct of generic properties of email IDs has a significant positive influence on the sender's adoption of bMail IDs ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H1 is supported.

2) **Test of Hypothesis H2.** The construct of prescriptive properties of bMail IDs has a significant positive influence on gaining individual trust as sender ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H2 is supported.

3) **Test of Hypothesis H3.** The senders' individual trust gained through bMail IDs has a significant positive influence on senders' adoption of bMail IDs ($p < 0.001$ in Table 4, and South Korea and USA in Table 5; and $p < 0.01$ in China in Table 5); thus, Hypothesis H3 is supported.

4) **Test of Hypothesis H4**. The changeability of the platform's member IDs in collaboration with the bMail ID service has a significant positive influence on senders' adoption of bMail IDs ($p < 0.001$ in Table 4 and the USA and China in Table 5; and $p < 0.01$ in South Korea in Table 5); thus, Hypothesis H4 is supported.

5) **Test of Hypothesis H5**. The construct of social properties of email IDs has a significant positive influence on the receivers' adoption of bMail IDs ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H5 is supported.

6) **Test of Hypothesis H6.** The construct of social properties of bMail IDs has a significant positive influence on building social trust as receiver ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H6 is supported.

7) **Test of Hypothesis H7.** The social trust established through the propagation of bMail IDs

has a significant positive influence on receivers' adoption of bMail IDs ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H7 is supported.

8) **Test of Hypothesis H8.** Both the sender's perspective and the receiver's perspective of adopting bMail IDs have a significant positive influence on users' synergistic adoption of bMail IDs ($p < 0.001$ in Tables 4 and 5); thus, Hypothesis H8 is supported.

According to the above test results of the eight hypotheses, we conclude that the netizens in these three countries are willing to adopt the properties of email IDs and bMail IDs to gain individual trust as senders and to build social trust as receivers. Based on this guideline for users' adoption, the bMail ID system can be designed according to these needs.

Table 4. Summary of SEM Results (Pooled Data)

Independent Variables	**Dependent Variables**				
	Sender-side Effect		**Receiver-side Effect**		**Synergy Effect**
	Sender's Adoption of bMail ID	Gain Individual Trust as Sender	Receiver's Adoption of bMail ID	Build Social Trust as Receiver	User's Adoption of bMail ID
Properties of eMail ID (H1)	**0.227*** **				
Properties of bMail ID (H2)		**0.652*** **			
Gain Individual Trust as Sender (H3)	**0.310*** **				
Resilience of Platform (H4)	**0.223*** **				
Social Properties of eMail ID (H5)			**0.184*** **		
Social Properties of bMail ID (H6)				**0.692*** **	
Build Social Trust as Receiver (H7)			**0.483*** **		
Sender's Adoption of bMail ID (H8)					**0.348*** **
Receiver's Adoption of bMail ID (H8)					**0.493*** **
Age	-0.038		-0.066*		
Sex	0.203***		0.217***		
International Interactions	0.030		0.038		
$\mathbf{R^2}$	0.402	0.426	0.354	0.479	0.590

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 5. Test of Hypotheses (Three Countries)

Structure Path	**Path Coefficient (p-value)**		
	South Korea	**USA**	**China**
H1: Properties of eMail ID → Sender's adoption of bMail ID	**0.263*** **	**0.284*** **	**0.178*** **
H2: Properties of bMail ID → Gain Individual Trust as Sender	**0.594*** **	**0.719*** **	**0.547*** **
H3: Gain Individual Trust as Sender → Sender's Adoption of bMail ID	**0.321*** **	**0.280*** **	**0.181** **
H4: Resilience of Platform → Sender's Adoption of bMail ID	**0.160** **	**0.245*** **	**0.234*** **

H5: Social Properties of eMail ID → Receiver's Adoption of bMail ID		**0.187*****	**0.284*****	**0.291*****
H6: Social Properties of bMail ID → Build Social Trust as Receiver		**0.749*****	**0.728*****	**0.519*****
H7: Build Social Trust as Receiver → Receiver's Adoption of bMail ID		**0.506*****	**0.439*****	**0.302*****
H8: Sender's Adoption of bMail ID → User's Adoption of bMail ID		**0.262*****	**0.312*****	**0.243*****
H8: Receiver's Adoption of bMail ID → User's Adoption of bMail ID		**0.598*****	**0.568*****	**0.374*****
Control Variables	Age → Sender's Adoption of bMail ID	0.052	0.014	-0.006
	Age → Receiver's Adoption of bMail ID	0.011	-0.094*	0.053
	Sex → Sender's Adoption of bMail ID	0.337***	0.149*	0.138
	Sex → Receiver's Adoption of bMail ID	0.189*	0.284***	0.053
	Internationalization → Sender's Adoption of bMail ID	-0.077	0.059	0.120
	Internationalization → Receiver's Adoption of bMail ID	0.013	0.024	0.033
R^2	Sender adoption of bMail ID	0.309	0.526	0.233
	Gain Individual Trust as Sender	0.353	0.517	0.299
	Receiver Adoption of bMail ID	0.336	0.470	0.253
	Build Social Trust as Receiver	0.561	0.530	0.269
	User's Adoption of bMail ID	0.635	0.685	0.277

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

6.2 Country Effects on Trust Building

Although we have validated that the bMail ID system is significantly adoptable in all three countries, Multigroup Analysis (PLS-MGA) was conducted to perform pairwise comparisons of constructs to explore differences in the effect levels of properties in the trust-building process between countries. According to the results in Table 6, respondents in the USA have a higher preference than those in China for establishing both sender-side individual trust and receiver-side social trust infrastructure. Note that the comparisons between South Korea and other countries are either not significant or the invariance condition is not established (denoted as "Not comparable") according to the MICOM (Measurement Invariance of Composite Models) criteria, as reported in Appendix Table C4 (Henseler et al., 2016; Luo et al., 2020).

Table 6. Country Differences in Establishing Individual and Social Trust

Structural Path	**Path Coefficient Difference (p-value)**		
	S. Korea vs. China	USA vs. S. Korea	USA vs. China
Properties of bMail ID → Gain Individual Trust as Sender	0.047 (0.521)	0.125* (0.012) Not comparable	**0.172*** (0.000)**
Social Properties of bMail ID → Build Social Trust as Receiver	0.230*** (0 .000) Not comparable	0.021 (0.323) Not comparable	**0.210*** (0.000)**

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

To explain the reasons for the country differences, we examine the role of cultural factors and the penetration levels of alternative telecommunication systems beyond email services in the three countries.

6.2.1 Effect of National Cultures on Trust Building

Previous studies indicated that national culture influences privacy concerns and trust formation (Kim et al., 2023) as well as the adoption and diffusion of information technologies (Srite & Karahanna, 2006). Accordingly, we examine the effect of cultural differences on the adoption of the bMail ID system. Hofstede's cultural dimensions theory (Hofstede, 1980; Hofstede et al., 2010) is widely used to characterize national cultures in six dimensions. In the context of bMail ID, we focus on two dimensions that are particularly relevant: *Individualism* (the degree to which people value personal autonomy over group membership) and *Power Distance* (the degree to which unequal power distribution is accepted). We contrast the implications of cross-country differences when these cultural dimension scores differ substantially across countries.

Effect of Individualism on Gaining Individual Trust as Sender. The individualism scores in the USA (91) are significantly higher than those in China (20) and South Korea (18) (Source: Hofstede et al., 2010). This high level of individualism reflects a stronger preference for directly gaining individual trust as a sender. According to the results reported in Table 6, the sender-side effect of the *'Properties of bMail ID'* on '*Gain Individual Trust as Sender'* in the USA has a significantly higher coefficient than that in China (0.172, $p < 0.001$) and South Korea (0.125, $p < 0.05$, although measurement invariance is not established). This result demonstrates that email senders in the high-individualism cultures, such as the USA, place a significantly stronger emphasis on individual trust-building mechanisms than those in lower-individualism cultures, such as China and South Korea. This is consistent with the cultural effect on trust in the context of e-commerce (Hallikainen & Laukkanen, 2018).

Implication on Adoption of Anonymity. High individualism is also associated with greater sensitivity to privacy protection via anonymity. According to the impact of anonymity on '*Gain Individual Trust as Sender'*, it is very interesting to find that anonymity is significant only in the USA (0.434, $p < 0.001$), whereas it is not significant in China (0.012, $p = 0.924$) and South Korea (0.162, $p = 0.270$) (as noted in Table 7). However, recognizability is found to be significantly important in all three countries

Effect of Power Distance on Business-Driven bMail ID. The power distance score in China (80) is significantly higher than the scores in the USA (40) and South Korea (60) (Source: Hofstede et al., 2010). In the context of bMail ID adoption, we interpret the high power distance score as a reflection of a stronger expectation of institutional trust mechanisms for building social trust, rather than of a business-driven trust mechanism such as the bMail ID system. According to the results reported in Table 6, the receiver-side effect of the '*Social Properties of bMail ID'* on *'Build Social Trust as Receiver'* in China has a significantly lower coefficient than those in the USA (0.210, $p <$ 0.001) and South Korea (0.023, p < 0.001, although measurement invariance is not established).

Effect of Power Distance on the Preference of bMail Certification Authority Type. We argue that the gap in power distance score might reflect the preferred types of Certification Authority (CA) for bMail identification. In China, 66% of respondents prefer governmental institutions as CA, while in the USA, only 48% of respondents prefer governmental institutions as CA (Table 8). In the USA, banks are preferred more as CA than telcos, possibly because telcos in the USA allow anonymous phone numbers while not in China and South Korea. The detailed roles of CAs are described in the next section.

Table 7. Variable-level Effect of eMail and bMail ID Adoption

eMail ID Properties	Sender Side	Receiver Side

	S. Korea	USA	China	S. Korea	USA	China
	eMail ID Adoption			eMail ID Adoption		
Communicability	0.309	0.171^{*}	0.028	0.318	0.395^{***}	0.520^{***}
Global Reach	0.169	0.321^{***}	0.289^{*}	-0.654^{*}	0.299^{*}	0.342^{*}
Memorability	0.140	0.286^{***}	0.446^{***}	0.851^{**}	0.265^{*}	0.171
Anonymity	0.162	0.434^{***}	0.012	--	--	--
Recognizability	0.719^{***}	0.402^{***}	0.616^{***}	0.603^{***}	0.312^{***}	0.354^{**}
bMail ID Properties	Gaining Individual Trust by bMail ID Adoption			Gaining Social Trust by bMail ID Adoption		
Identifiability	0.204^{*}	0.495^{***}	0.393^{***}	0.626^{***}	0.399^{***}	0.427^{***}
Verifiability	0.394^{***}	0.303^{**}	0.472^{***}	0.510^{***}	0.422^{***}	0.444^{***}
Sustainability	0.625^{***}	0.407^{***}	0.496^{***}	0.037	0.374^{***}	0.483^{***}

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 8. Preference of Certification Authority Types in Three Countries

Certification Authority	South Korea	USA	China
Telco Company	66.93%	36.59%	72.92%
Bank	60.21%	64.55%	66.51%
Trustable Government Agency	66.93%	48.86%	66.51%

6.2.2 Effect of Alternative Domestic Telecommunication Infrastructure

Besides the cultural differences, telecommunication infrastructures also shape trust-building pathways. Recall that the effect of recognizability on the adoption of the eMail ID system was consistently significant across all three countries, both from sender and receiver perspectives (Table 7). Nevertheless, the levels between countries differ substantially depending on the penetration level of messenger services as an alternative telecommunication channel. In the USA, the impact level of recognizability is 0.402, which is lower than that in China (0.616). We argue that this difference is attributable to the dependency on regional messenger services in China. In the USA, email is a deeply embedded communication channel in professional and personal contexts. However, in China, messenger services such as WeChat and QQ are so popular that they have penetrated both domestic social and business interactions through their own identity verification mechanisms. Thus, the domestic reliance on email-based communication is accordingly reduced in China. Given the dominance of instant messenger services that fulfill the needs of domestic communication, email is relatively used more for international communication.

Consequently, for Chinese senders, high *recognizability* and *global reach* become critical properties for efficient and secure international interactions, whereas *communicability* and *anonymity* of sending email IDs are not primary concerns for domestic interactions (Table 7).

Similarly, in South Korea, the impact level of *recognizability* (0.719) is notably significant, whereas the other four email properties lack statistical significance. This pattern results from the extensive adoption of KakaoTalk for domestic personal interactions (KOISRA, 2023), diminishing the relative importance of domestic email communication. Therefore, South Korean users prioritize recognizability in email communication for serious business communication (Table 7).

Consequently, these findings underscore the importance of jointly considering cultural values and local alternative telecommunication infrastructures in shaping the adoption intentions of bMail IDs across different countries. However, these differences do not mitigate the need for adopting bMail IDs to use as member IDs for diverse e-commerce and e-service platforms, as we validated through the tests of eight hypotheses.

7. ARCHITECTURE OF bMail ID AND bMember PLATFORMS

Based on the five validated features of email (communicability, global reach, memorability, anonymity, and recognizability) and the four validated features of bMail (identifiability, verifiability, sustainability, and changeability), we conceptually design the architecture of the bMail ID infrastructure for online platforms. From the membership management perspective, this architecture can be applied to diverse online platforms, which we denote as Bright Member platforms (in short, bMember platforms). When the platform is an e-commerce platform, we refer to it as a Bright E-Commerce (in short, bCommerce) platform.

The entities of the bMail ID infrastructure include *bMail users*, *bMail ID Service Provider* (*bMail ISP*), *bMember platforms,* and *bMail ID Certification Authority* (*bCA*). The basic architecture of the bMail ID platform is depicted in Figure 3, which illustrates both the identification process of bMail IDs and the verification process of genuine senders against phishing attempts. The features of registering and updating bMembers are depicted in the lower section. The functions of these entities are summarized as follows, without specifying their technical implementations.

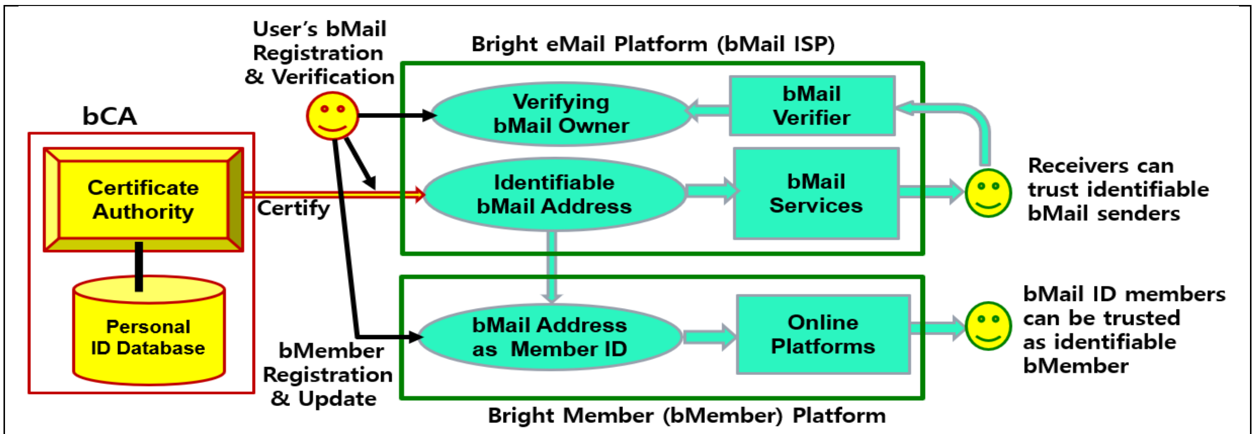

Figure 3. Architecture of the bMail ID and bMember Platforms

bMail Users. Both individuals and business users may register a bMail account at a bMail ISP. Individual users may use their conventional emails for private communication, but may also use bMail IDs for their official and business communications and interactions with online platforms. Business users are likely to use bMail accounts for their trustworthy business communications. The process of bMail ID registration and use is described in four steps.

1) Registration at a bMail ISP

① To register a bMail account, users should voluntarily permit the assurance of identifiability of their genuine IDs based on permitted personal data (such as name, affiliation, address, and phone number) that are already stored in a selected bCA (such as a telco or bank).

② Users should also provide a verification communication channel to the bMail ISP so that it can be used for verification upon request by legitimate recipients.

③ Upon application and validation of users, the bMail ISP issues a bMail ID account that is based on the bMail domain name such as bgmail.com, either as a real-named bMail ID (such as efraim.turban.hawaii@bgmail.com) or an anonymous bMail ID (such as 19905217@bgmail.com).

2) Using the registered bMail address as an ID

① Send bMail messages through the bMail ISP using the bMail ID

② Register memberships on online platforms using the bMail ID

3) Upon a valid request for identification from a receiver through the bMail ISP

① The bMail ISP reviews the validity of the receiver's request to identify the sender. If the request is valid, the bMail ISP provides the sender's permitted identification data to the receiver so that the sender is deemed trustworthy.

② The bMail ISP notifies the sender of the identification event.

4) Upon a valid request for verification from a receiver through the bMail ISP

① The bMail ISP reviews the validity of the verification request. If the request is valid, the bMail ISP forwards the verification request to the genuine sender through a verification channel.

② The genuine sender confirms to the receiver, through the bMail ISP, whether the delivered bMail was sent by the genuine sender or not.

bMail ID Service Providers. The bMail ISPs must support the four prescriptive features (identifiability, verifiability, sustainability, and changeability) of bMail IDs while protecting users' privacy without causing an intolerable transactional burden. Traditional email service providers are candidates to become bMail ISPs to upgrade their services and expand their business scope as bMember ID managers. The tasks that bMail ISPs should perform are described as follows.

(1) Identifiability Services:

1) Agreements with Certification Authorities

① Establish collaborative agreements with third-party certification authorities (bCAs), such as telcos or banks, by partnering with those that already possess validated genuine personal identification data.

② bMail ISPs do not copy or store users' personal data in their own storage to avoid potential risks of accidental disclosure.

2) Securing bMail Domain Name

① bMail ISPs should register bMail domain names in a recognizable manner, such as bgmail.com or b.gmail.com.

② Ensure the adoption of authentication protocols that prevent spoofing during email message transmission, as most email service providers already do.

③ Publicize the registered bMail domain names through an established bMail domain portal.

3) Agreement of the Identification Service at bMail User's Registration

① Require bMail users to agree to the identification permissions and select a bCA.

4) Identification Service for bMail Receivers

① Upon a valid request for sender identification from bMail receivers, notify the request to the bMail sender for their review and permission for identification.

② If the bMail sender agrees, request the associated bCA to issue the identity certificate and deliver it to the requesting receiver, and confirm the issuance with the sender.

(2) Verification Services:

1) Agreement of Verification Service at bMail User's Registration

① Acquire verification communication channels from registered bMail users, such as backup email addresses, messenger IDs, and/or phone numbers.

② Assure that the verification channels remain effective through periodic validation.

2) Verification Service for bMail Receivers

① Upon a valid request for sender verification from bMail receivers, notify the request to the bMail sender for their permission to proceed with verification.

② If the bMail sender agrees, deliver the verification message to the requesting receiver and confirm the verification with the sender.

(3) Sustainable Lifelong Services: The bMail account service should be sustainably continued for the lifetime of users unless a user commits misconduct and loses legitimacy. The bMail ISPs should announce that their bMail IDs are guaranteed for lifelong service to make them more trustworthy.

(4) Facilitate a Secure ID Change Protocol with bMember Platforms: To allow the valid change of email IDs to bMail IDs securely, bMail ISPs should establish a protocol that can be adopted by bMember platforms in order to prevent malicious changes of member IDs.

(5) Monitor Statistics. bMail ISPs should monitor the propagation statistics of bMail IDs and measure the effectiveness of building social trust.

bMember Platforms. bMember platforms, such as bCommerce platforms, need to retain bMember IDs for lifelong services and adopt a secure protocol to support the legitimate change of their customers' IDs to bMember IDs when it becomes necessary.

1) Register bMembers: Recognize bMembers as identifiable and verifiable lifelong members in their customer relationship management systems.

2) Establish a secure protocol for changing member IDs: Allow the legitimate change of IDs when the email ID has become ineffective or when users want to change to bMail IDs. To allow the update of their member IDs by associating the new IDs seamlessly with the current customer records, it is necessary to establish a secure protocol with bMail ISPs to prevent the risk of fraudulent changes.

E-commerce platforms need to adopt bMail ID membership to upgrade the trust level of customers and retain sustainable customer relationships. As e-commerce platforms become bCommerce platforms, they will be regarded as highly socially responsible organizations, facilitating customers' individual trust and enhancing the social reputation of the bCommerce platform. Likewise, social networks can adopt the bMember ID system, reconciling identifiability and anonymity and preventing fake news generated by fake IDs.

bMail ID Certification Authority (bCA). Certification authorities for bMail IDs certify the identifiability of bMail accounts. The role of bCAs may be performed by organizations that keep genuine identity data by the nature of their business, such as telecommunications companies, banks, credit card companies, or governmental institutions. Users' preferences among these alternatives in three countries are compared in Table 8. As discussed earlier, in the USA, banks are more highly favored than telcos and government institutions. On the other hand, in China and South Korea, both telcos and banks are preferred, as well as governmental institutions. The bCA service can be positioned as a new customer service to attract more customers. bMail ISPs need to establish partnerships with a wide range of collaborating bCAs to cover potential customers in diverse settings. The bCAs certify the identifiability of users even if they may use pseudonyms in their bMail IDs.

Business Models. Based on the consortium of bMail ISPs, bMember platforms, and bCAs, the technologies for implementing the conceptual architecture of the bMail infrastructure can be developed. The entities in the consortium can pursue new business models collaboratively as described in the Practical Implications section. Email service providers can extend their business as globally trustworthy and safe email and ID service providers; telcos and banks can offer extended identification services to their clients, thereby attracting their legitimate customers; and eCommerce and eService platforms can retain their trustworthy customers sustainably. Their global collective efforts will establish an infrastructure for individual and social trust while enhancing the safety of global cyberspace.

8. CONCLUSION AND DISCUSSION

8.1 Summary

This research aims to identify the prescriptive properties of the bMail ID infrastructure and design the necessary features of the bMail-based ID system that augment both individual and social trust without infringing on privacy. As the guidance of proactive prescriptive design is necessary for the development of the bMail ID infrastructure that will fulfill the trust and safety of cyberspace, an *a priori* artifact is proposed according to the framework of design science research guidelines as summarized in Table 9 (Hevner et al., 2004). Thus, we identified the prescriptive properties of the bMail-based ID system and associated them with the design goals of enhancing individual trust for senders and establishing social trust for receivers of emails.

To postulate the research model, we considered the five generic advantages of email-based ID systems (communicability, global reach, memorability, optional anonymity, and optional recognizability) and identified four limitations that the current email-based ID systems do not support (identifiability, verifiability, sustainability, and changeability) from both sender and receiver perspectives. These limitations derive into the necessary properties of the bMail ID infrastructure. The five generic properties of email-based ID system and the four prescriptive properties of the bMail ID system formed the structural equation model that associates them with the goals of senders' individual trust and receivers' social trust. These complementary goals will jointly motivate the users' adoption of the bMail ID system. The research model was tested and empirically supported using cross-country survey data from South Korea, the United States, and China.

We also examined the effects of national culture and the penetration level of messenger services on the adoption of the bMail ID system. The effect of individualism is particularly examined because societies with high individualism, such as the USA, prefer privacy protection through anonymity while identifiability is guaranteed. The bMail-based ID system resolves the conflicting need for identifiability and anonymity by adopting the notion of identifiable anonymity, which allows email names to be anonymous while the bMail domain names assure identifiability in alignment with the vision of the Bright Origin (Lee, 2015).

Based on our findings, we designed the architecture of the bMail ID infrastructure by identifying the roles of key stakeholders: bMail ID users, bMail ID service providers, bMember platforms, and bMail ID certification authorities such as telcos and banks. Through the collaborative business models of this infrastructure, more secure management of bMember IDs and bCommerce platforms will become possible, thereby contributing to a preventively safer cyberspace. Nevertheless, identifiability alone cannot assure the accountability.

Table 9. Design Science Research Guidelines

DSR Guidelines	How this study addresses the guideline
Problem Relevance	This study identifies the limitations of conventional email-based online ID systems and articulates the need for an enhanced ID system to establish individual trust for senders and social trust infrastructure for receivers in online communications and membership management.
Design as	This study designs the bMail ID system that meets the requirements of an identifiable,

an Artifact	verifiable, sustainable, and resilient ID system while offering the option of anonymity, and is applicable to member management across various e-commerce and online service platforms.
Design Evaluation	The proposed properties of the bMail ID system are evaluated empirically through survey data collected from business-experienced netizens in South Korea, the United States, and China, validating both the four descriptive and the five prescriptive design features.
Research Contribution	This study contributes by integrating descriptive properties of existing email IDs with prescriptive design principles to propose a new trustworthy online bMail ID artifact that supports trust formation at both individual and social levels.
Research Rigor	This study applies structural equation modeling to analyze survey data, examines mediating trust mechanisms, and investigates country effects, incorporating cultural factors and the role of messenger systems as an alternative online ID mechanism.
Design as a Search Process	The architecture of the bMail ID system is systematically designed to satisfy identified requirements, with explicit modeling of entities, roles, and interactions, reflecting a structured search for an effective design solution.
Communication of Research	The paper presents both the technical design of the bMail ID system and its practical implications for customer management, trust building, and identity governance in e-commerce and online service contexts.

8.2 Reconciliation with Freedom of Expression

We discuss whether the voluntary propagation of identifiability via the bMail ID infrastructure may inhibit the freedom of anonymous expression that is necessary for whistleblowers. Although the dark web may be innocently used for anonymous freedom of speech, it should not be abused to contaminate society as an instrument of global cybercrime.

On the contrary, the purpose of the identifiable anonymity principle adopted in the bMail ID infrastructure is to build a voluntary cyberspace that establishes trustworthiness and safety for exchanging emails and conducting social activities among legitimate users. This approach resembles building a credible society that uses credit cards while anonymous cash is still available. In this regard, the cyberspace of the bMail ID infrastructure does not prohibit a parallel cyberspace that enables freedom of anonymous expression, and both can coexist, balancing different roles in society.

8.3 Theoretical Implications

This paper makes several theoretical contributions. First, this study introduces a proactive paradigm in cybersecurity through the bMail ID artifact. Unlike traditional methods that defensively respond to cybercrimes, this study proactively mitigates cybersecurity risks at their origin by applying the identifiable anonymity principle. By adopting bMail IDs for members of platforms, e-commerce platforms can mitigate fake reviews generated by fake IDs, and social network platforms can likewise mitigate fake news generated by anonymous authors or automated software.

Second, this research theoretically identifies the necessary features of online IDs, such as identifiability, verifiability, sustainability, and changeability, by extending the design dimensions of digital identity. Our empirical findings from surveys conducted in South Korea, the United States, and China validate the effectiveness of these features.

Third, we enrich the design science literature by integrating both sender and receiver perspectives in digital identity adoption. By examining the synergistic relationship between sender and receiver adoption behaviors, our study enhances understanding of the formation of individual and social trust in digital identity systems.

Fourth, this research advances cross-cultural information systems literature by examining the effects of culture–particularly individualism and power distance–on the choice between anonymity for privacy protection and real-named email for enhancing recognizability and trust. The effect of the high penetration of regional messenger services is also examined to contextualize the need for bMail IDs.

Finally, our study exemplifies an integration of design science and prescriptive behavioral science methodologies. Initially, we analyzed the limitations in current email systems and designed the necessary features to address these gaps. Subsequently, we validated the features using surveys of potential users' behavior. By combining rigorous design with prescriptive validation, this hybrid approach provides a comprehensive and effective framework for future cybersecurity infrastructure design.

8.4 Practical Implications

The findings from this study offer valuable guidance for practitioners in the field. First, our research advocates a proactive cybersecurity approach, emphasizing identifiability and verifiability as deterrents to cybercrimes. By implementing the identifiable anonymity concept, the proposed bMail ID system would substantially mitigate criminal activities conducted under anonymity by distinguishing voluntary identifiable users from anonymous malicious actors. Email service providers can extend their business models by providing trustworthy bMail services and online IDs.

Second, as e-commerce platforms adopt bMembers who use bMail IDs as their member IDs, they can upgrade the trust level of their members, preventing fake behaviors and fake reviews on their platforms and making them socially trustworthy as bCommerce platforms. Social network platforms can adopt bMail IDs as their member IDs, preventing the spread of fake news without infringing on users' privacy through identifiable anonymity. Both bCommerce and bSocial network platforms can enhance their reputations for social responsibility.

Third, current online platforms do not allow users to change their member IDs even when the associated email address is no longer in service. Therefore, platforms may lose their current customers whose email addresses are suspended. Consequently, online platforms need a secure protocol for legitimately changing member IDs in collaboration with bMail ID service providers. Since bMail IDs will be effective for a lifetime, this approach will enable platforms to retain contact with customers throughout their lifetime.

Fourth, to ensure the identifiability of bMail IDs, bMail ID service providers should associate their users with third-party bCAs such as telcos and banks, which possess users' verified identity data by the nature of their business. Since certification services can add benefits to their customers, this represents a business opportunity for these bCAs.

8.5 Limitations and Future Research

The design of the bMail ID infrastructure in this paper is at the stage of conceptual design with a pilot study. The concept will become more specified depending upon the characteristics of the adopted technologies for implementation. Therefore, specific technical and business model research needs to be conducted to implement the processes of certification, identification, verification, and the protocol for changing member IDs. We will also need to test the empirical behavioral changes under identifiability.

Our survey questionnaires contained many items that might have increased respondents' impatience. Thus, we employed a single-item measure to express the key concept for each variable. Although we believe the current questionnaires clearly represent the concepts of the variables and constructs, future studies may consider different ways of expressing potentially ambiguous concepts within the same survey format to further improve the reliability and validity of the measures.

REFERENCES

Allen, C. (2016). *The path to self-sovereign identity*. Life With Alacrity. https://www.lifewithalacrity.com/article/the-path-to-self-sovereign-identity/

An, S., Jiang, L., & Eck, T. (2026). Are dark tourism experiences memorable? Examining mindfulness and memorable tourism experiences in dark tourism. *Humanities and Social Sciences Communications*, 13, 88. https://doi.org/10.1057/s41599-025-06394-3

Chesney, T., & Su, D. K. S. (2010). The impact of anonymity on weblog credibility. *International Journal of Human-Computer Studies*, *68*(10), 710–718. https://doi.org/10.1016/j.ijhcs.2010.06.001

Cybersecurity and Infrastructure Security Agency. (2022). *4 things you can do to keep yourself cyber safe*. https://www.cisa.gov/news-events/news/4-things-you-can-do-keep-yourself-cyber-safe

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, *13*(3), 319–340. https://doi.org/10.2307/249008

Day, S. (2024). *Embracing sustainability in e-commerce: A path to a greener future*. DoubleDome.

https://www.doubledome.com/blog/ecommerce-development/ecommerce-sustainability-path-greener-future/
Dellarocas, C. (2003). The digitization of word of mouth: Promise and challenges of online feedback mechanisms. *Management Science*, *49*(10), 1407–1424. https://doi.org/10.1287/mnsc.49.10.1407.17308
Dürscheid, C., & Frehner, C. (2013). Email communication. In S. C. Herring, D. Stein, & T. Virtanen (Eds.), *Pragmatics of computer-mediated communication* (pp. 35–54). De Gruyter Mouton. https://doi.org/10.1515/9783110214468.35
Federal Trade Commission. (2024). *Consumer sentinel network data book 2024*. https://www.ftc.gov/system/files/ftc_gov/pdf/csn-annual-data-book-2024.pdf
Feigenbaum, J., Jaggard, A. D., & Wright, R. N. (2014). Open vs. closed systems for accountability. In *Proceedings of the 2014 Symposium and Bootcamp on the Science of Security* (pp. 1–11). https://doi.org/10.1145/2600176.2600179
Flacandji, M., & Krey, N. (2020). Remembering shopping experiences: The shopping experience memory scale. *Journal of Business Research*, *107*, 279–289. https://doi.org/10.1016/j.jbusres.2018.10.039
Forman, C., Ghose, A., & Wiesenfeld, B. (2008). Examining the relationship between reviews and sales: The role of reviewer identity disclosure in electronic markets. *Information Systems Research*, *19*(3), 291–313. https://doi.org/10.1287/isre.1080.0193
Gefen, D., Karahanna, E., & Straub, D.W. (2003). Trust and TAM in online shopping: An integrated model. *MIS Quarterly*, *27*(1), 51. https://doi.org/10.2307/30036519
Gimpel, H., Heger, S., Olenberger, C., & Utz, L. (2021). The effectiveness of social norms in fighting fake news on social media. *Journal of Management Information Systems*, *38*(1), 196–221. https://doi.org/10.1080/07421222.2021.1870389
Haak, L. L., Fenner, M., Paglione, L., Pentz, E., & Ratner, H. (2012). ORCID: A system to uniquely identify researchers. *Learned Publishing*, *25*(4), 259–264. https://doi.org/10.1087/20120404
Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2022). *A primer on partial least squares structural equation modeling (PLS-SEM)* (3rd ed.). Sage.
Hallikainen, H., & Laukkanen, T. (2018). National culture and consumer trust in e-commerce. *International Journal of Information Management*, 38(1), 97–106. https://doi.org/10.1016/j.ijinfomgt.2017.07.002
He, S., Hollenbeck, B., & Proserpio, D. (2022). The market for fake reviews. *Marketing Science*, *41*(5), 896-921. https://doi.org/10.1287/mksc.2022.1353
Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. *International Marketing Review*, *33*(3), 405–431. https://doi.org/10.1108/IMR-09-2014-0304
Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, *28*(1), 75–105.
Hofstede, G. (1980). *Culture's consequences: International differences in work-related values*. Sage.
Hofstede, G., Hofstede, G. J., & Minkov, M. (2010). *Cultures and organizations: Software of the mind* (3rd ed.). McGraw-Hill.
Hoory, L. (2023). *The state of workplace communication*. Forbes Advisor. https://www.forbes.com/advisor/business/digital-communication-workplace/
Inquivix. (2023). *How to find KakaoTalk ID and why it matters in Korea's #1 app*.

https://inquivix.com/how-to-find-kakaotalk-id/

Javelin Strategy & Research. (2024). *Identity fraud flourishes amid consumers' growing digital presence, costing them time and money*. GlobeNewswire. https://www.globenewswire.com/news-release/2024/04/10/2860874/0/en/Identity-Fraud-Flourishes-Amid-Consumers-Growing-Digital-Presence-Costing-Them-Time-and-Money.html

Kim, D., Valacich, J. S., Jenkins, J. L., Wilson, D. W., Kumar, M., & Weisgarber, P. (2026). Are you, you? Seamlessly fighting identity fraud with keystroke dynamics. *Information Systems Research*, *0*(0). https://doi.org/10.1287/isre.2023.0088

Kim, H. W., & Kankanhalli, A. (2009). Investigating user resistance to information systems implementation: A status quo bias perspective. *MIS Quarterly*, *33*(3), 567–582. https://doi.org/10.2307/20650309

Kim, Y., Srite, M., & Zhao, H. (2023). Trust antecedents in online reviews across national cultures. *Decision Support Systems*, *173*, 113998. https://doi.org/10.1016/j.dss.2023.113998

Kock, N. (2015). Common method bias in PLS-SEM: A full collinearity assessment approach. *International Journal of e-Collaboration*, *11*(4), 1–10. https://doi.org/10.4018/ijec.2015100101

KOISRA. (2023). *Messaging apps for business in Korea: Do's and don'ts*. https://www.koisra.co.kr/doing-business-in-korea/messaging-apps-for-business-in-korea/

Kshetri, N. (2017). Blockchain's roles in strengthening cybersecurity and protecting privacy. *Telecommunications Policy*, *41*(10), 1027–1038. https://doi.org/10.1016/j.telpol.2017.09.003

Lareki, A., Altuna, J., & Martínez-de-Morentin, J.I. (2023). Fake digital identity and cyberbullying. *Media, Culture & Society*, *45*(2), 338–353. https://doi.org/10.1177/01634437221126081

Lee, J., Cho, D., & Lim, G. G. (2018). Design and validation of the bright internet. *Journal of the Association for Information Systems*, *19*(2), 63-85. https://aisel.aisnet.org/jais/vol19/iss2/3

Lee, J. K. (2015). Guest editorial: Research framework for AIS grand vision of the bright ICT initiative. *MIS Quarterly*, *39*(2), iii–xii.

Liu, G., Gao, X., & Wang, H. (2021). An investigation of identity-account inconsistency in single sign-on. In *Proceedings of the Web Conference 2021* (pp. 105–117). https://doi.org/10.1145/3442381.3450085

Luo, X. (Robert), Li, H., Hu, Q., & Xu, H. (2020). Why individual employees commit malicious computer abuse: A routine activity theory perspective. *Journal of the Association for Information Systems*, *21*, 1552–1593. https://doi.org/10.17705/1jais.00646

MacKenzie, S. B., Podsakoff, P. M., & Podsakoff, N. P. (2011). Construct measurement and validation procedures in MIS and behavioral research: Integrating new and existing techniques. *MIS Quarterly*, *35*(2), 293–334. https://doi.org/10.2307/23044045

Malhotra, N. K., Kim, S. S., & Patil, A. (2006). Common method variance in IS research: A comparison of alternative approaches and a reanalysis of past research. *Management Science*, *52*(12), 1865–1883. https://doi.org/10.1287/mnsc.1060.0597

Markman, K. M., & Scott, C. R. (2005). Anonymous internet? Examining identity issues in email addresses. *Annual Meeting of the International Communication Association, New York, NY*.

McKnight, D. H., Choudhury, V., & Kacmar, C. (2002). Developing and validating trust measures for e-commerce: An integrative typology. *Information Systems Research*, *13*(3), 334–359. https://doi.org/10.1287/isre.13.3.334.81

Meta. (2025). *Meta Verified: get a verified blue check on Instagram, Facebook*. https://about.meta.com/technologies/meta-verified/

Microsoft. (1999). *Microsoft passport offers streamlined purchasing across leading web sites*. https://news.microsoft.com/source/1999/10/11/microsoft-passport-offers-streamlined-purchasing-across-leading-web-sites/

Morgan, R. M., & Hunt, S. D. (1994). The commitment-trust theory of relationship marketing. *Journal of Marketing*, *58*(3), 20–38. https://doi.org/10.1177/002224299405800302

Mueller, M. L., Park, Y., Lee, J., & Kim, T. Y. (2006). Digital identity: How users value the attributes of online identifiers. *Information Economics and Policy*, *18*(4), 405–422. https://doi.org/10.1016/j.infoecopol.2006.04.002

Naver. (2025). *Unable to change the ID and account owner*. https://help.naver.com/service/5640/contents/20861?lang=en&osType=COMMONOS

Paget, S. (2024). *Local consumer review survey 2024: Trends, behaviors, and platforms explored*. BrightLocal. https://www.brightlocal.com/research/local-consumer-review-survey-2024/

Pitt, J. (2025). *2025 Identity fraud study: Breaking barriers to innovation*. Javelin. https://javelinstrategy.com.

Pu, J., Chen, Y., Qiu, L., & Cheng, H. K. (2020). Does identity disclosure help or hurt user content generation? Social presence, inhibition, and displacement effects. *Information Systems Research*, *31*(2), 297–322. https://doi.org/10.1287/isre.2019.0885

Qian, B., Koh, T. K., & Zhang, X. (Michael). (2025). From anonymity to accountability: How virtual identity disclosure changes the quantity and quality of "likes." *Information Systems Research*, *36*(3), 1926–1937. https://doi.org/10.1287/isre.2020.0335

Roghanizad, M. M., & Bohns, V. K. (2017). Ask in person: You're less persuasive than you think over email. *Journal of Experimental Social Psychology*, *69*, 223–226. https://doi.org/10.1016/j.jesp.2016.10.002

Samtani, S., Chinn, R., Chen, H., & Nunamaker, J. F. (2017). Exploring emerging hacker assets and key hackers for proactive cyber threat intelligence. *Journal of Management Information Systems*, *34*(4), 1023–1053. https://doi.org/10.1080/07421222.2017.1394049

Sen, R., & Borle, S. (2015). Estimating the contextual risk of data breach: An empirical approach. *Journal of Management Information Systems*, *32*(2), 314–341. https://doi.org/10.1080/07421222.2015.1063315

Shen, K., Wang, C., Guo, M., Zheng, X., Lu, C., Liu, B., Zhao, Y., Hao, S., Duan, H., Pan, Q., & Yang, M. (2021). Weak links in authentication chains: A large-scale analysis of email sender spoofing attacks. In *Proceedings of the 30th USENIX Security Symposium (USENIX Security 21)* (pp. 3201–3217). https://www.usenix.org/conference/usenixsecurity21/presentation/shen-kaiwen

Shukla, A. D., & Goh, J. M. (2024). Fighting fake reviews: Authenticated anonymous reviews using identity verification. *Business Horizons*, *67*(1), 71–81. https://doi.org/10.1016/j.bushor.2023.08.002

Srite, M., & Karahanna, E. (2006). The role of espoused national cultural values in technology acceptance. *MIS Quarterly*, *30*(3), 679. https://doi.org/10.2307/25148745

Thomas, M., Goic, M. and Kalyanam, K. (2025). Long lags and large returns: Experimental evidence from advertising to businesses. *Management Science*, *71*(10), 8750-8766. https://doi.org/10.1287/mnsc.2023.02661

University of Minnesota. (2024). *Changes to UMN email and google workspace*. https://isss.umn.edu/node/4281

Vecchietti, G., Liyanaarachchi, G., & Viglia, G. (2025). Managing deepfakes with artificial intelligence: Introducing the business privacy calculus. *Journal of Business Research*, *186*,

115010. https://doi.org/10.1016/j.jbusres.2024.115010
Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, *27*(3), 425–478. https://doi.org/10.2307/30036540
Wu, T., Peng, C.-H., Sia, C. L., & Lu, Y. (2024). Website localization strategies to promote global e-commerce: The moderating role of individualism and collectivism. *MIS Quarterly*, 48(1), 31–66. https://doi.org/10.25300/MISQ/2022/15542
Wu, Z., Liu, G., Wu, J., & Tan, Y. (2024). Are neighbors alike? A semisupervised probabilistic collaborative learning model for online review spammers detection. *Information Systems Research*, *35*(4), 1565–1585. https://doi.org/10.1287/isre.2022.0047
X Corp. (2025). *How to get the blue checkmark on X.* X Help Center. https://help.x.com/en/managing-your-account/about-x-verified-accounts
Xu, J. J., Chen, D., Chau, M., Li, L., & Zheng, H. (2022). Peer-to-peer loan fraud detection: Constructing features from transaction data. *MIS Quarterly*, *46*(3), 1777–1792. https://doi.org/10.25300/MISQ/2022/16103
Yildiz, H., Ritter, C., Nguyen, L. T., Frech, B., Martinez, M. M., & Kupper, A. (2021). Connecting Self-Sovereign Identity with federated and user-centric identities via SAML integration. In *Proceedings of the 2021 IEEE Symposium on Computers and Communications (ISCC)* (pp. 1–7). https://doi.org/10.1109/ISCC53001.2021.9631453

APPENDIX A: Text Mining of Cybersecurity Incidents

To study the need for identifiability and verifiability in the context of cybersecurity incidents, we applied text mining techniques to quantify the frequency of four bMail property occurrences in the incident database. To this end, we analyzed 81,847 cybersecurity incident records from the Privacy Rights Clearinghouse (PRC) dataset, covering incidents from 2005 to 2025. Each incident report includes a textual description, date of breach, affected organization type, and occasionally the number of impacted individuals. Building on established text mining protocols in cybersecurity research (Samtani et al., 2017), we developed a comprehensive pattern-matching framework specifically designed to identify bMail feature requirements. This framework consists of three distinct phases:

Phase 1: Seed Term Development: We developed an initial taxonomy of security threats related to identity, phishing, sustainability, and changeability based on existing cybersecurity literature and incident reports. Core categories included fake identity attacks, phishing attempts, account compromise, email sustainability issues, and ID changeability on online platforms.

Phase 2: Pattern Recognition and Classification: Using regular expressions and natural language processing techniques, we adopted a keyword matching method to identify incidents corresponding to each threat category. Table A1 documents keyword examples for identifiability and verifiability to classify PRC incident narratives into the two bMail design properties. When multiple categories match a narrative, a primary label is assigned based on contextual weighting, while secondary labels are retained for robustness checks.

Table A1. Keywords Detected by Text mining

Variable	Keyword Examples
Identifiability	fake ID, fake identity, identity theft, unauthorized access, unauthorized user, impersonation, fraudulent account, fake account, stolen credentials, compromised account, unknown individual, unverified user, insider threat
Verifiability	phishing, spoof, deceptive, fraudulent email, suspicious email, fake website, misleading, scam, verification failure, cannot verify, unconfirmed source

Phase 3: Feature Mapping and Validation: Our pattern matching employed word-boundary detection and case-insensitive matching to ensure robust identification while minimizing false positives. From the PRC database, a total of 28,032 out of 81,847 incidents were classified as relevant to bMail feature requirements. The incidents relevant to identifiability and verifiability are summarized in Table A2. This mapping established direct empirical links between observed security incidents and proposed preventive measures. However, no cybersecurity incidents were found to be related to the sustainability and changeability of IDs.

Table A2: Text Mining Results on bMail Feature Requirements

	bMail Feature	Incident Count	Percentage
PRC Database	Identifiability	25,928	92.49%
	Verifiability	2,104	7.51%
	Total Classified	28,032	100.00%

APPENDIX B

Table B1. Details of Variables and Latent Constructs of Survey

7-point Likert scale [3: Definitely yes, 0: Neutral, -3: Definitely no]	
We restricted participants to those with work experience.	
Variables	**Items**
Communicability (V1)	I need to use email services for my daily life and business activities.
Global Reach (V2)	I need to use email services for my cross-border communications and business activities.
Memorability (V3)	I prefer to use an email address as my online member ID because I can easily remember my email address as my member ID.
Anonymity (V4)	I prefer to pseudonymize my email address to hide my real name.
Recognizability (V5)	I prefer to send my emails using a recognizable email name so that the receivers can recognize my name without additional information.
Identifiability (V6)	To be trusted by unknown receivers (even though I may not disclose my real name in the email name), I am willing to permit the provision of my generic identity data when it is requested by valid recipients.
Verifiability (V7)	I am willing to register backup communication channels so that innocent receivers who face phishing attacks from my email ID can contact me to verify the truth through these backup communication channels.
Sustainability (V8)	I believe that my lifelong email address (certified by the brand of services) will be trusted more than an ordinary email address by unknown recipients of my emails.
Changeability (V9)	When the old email ID is terminated, I believe that e-commerce and e-service platforms should allow users to change to a new ID through a secure change protocol.
Gain Individual Trust (Mediating Goal: MD)	• It is necessary to upgrade the current email-based ID system so that innocent senders can be trusted by unknown recipients. • It is necessary to upgrade the current email-based ID system so that innocent senders can be trusted by unknown counterparts in e-commerce and e-service platforms.
Communicability with Senders (VS1)	I prefer to use an email address as my online member ID because my registered e-commerce and e-service platforms can easily contact me through my email-based ID.
Global visibility to Senders (VS2)	I prefer to use an email address as my online member ID because my registered cross-border e-commerce and e-service platforms can easily contact me through my email-based ID.
Being Remembered by Senders (VS3)	I prefer to use an email address as my online member ID because my counterparts in e-commerce can recall my email-based ID more easily using their records of exchanged emails with me than any other ID.
Recognizable Sender (VS5)	I prefer to receive emails whose senders' real names are recognizable.
Identifiable Sender (VS6)	I would trust the received email highly if the identifiability of sender's real name is certified by a trustworthy third party (such as a telco or a bank).
Verifiable Sender (VS7)	I need a new type of email facility that enables receivers to respond back to the true owner of a suspicious email address through backup communication channels even though this information itself is not disclosed to me.
Sustainable Sender (VS8)	I would trust a received email highly if it used a lifelong email name (certified by a brand of services) even though the sender was not known to me in advance.
Build Social Trust (Mediating Goal: MDS)	• It is necessary to upgrade the current email-based ID system so that emails received from unknown innocent people can be more trusted. • It is necessary to upgrade the current email-based ID system so that unknown innocent counterparts in e-commerce and e-services can be trusted more.

Sender's Adoption of bMail ID	I want to have a bMail-based ID to send my important innocent emails to unknown receivers and to use as my member ID in the e-commerce and e-service platforms.
Receiver's adoption of bMail ID	I believe that the innocent netizens of society need to collectively adopt the bMail-based ID so that receivers of bMail can highly trust innocent senders' emails even though they were not known before.
User's adoption of bMail ID	I want to adopt the bMail-based ID with higher confidence if bMail services are widely adopted by the innocent netizens.

Table B2. Demographics of the Data

		Total Data, N=1,248	South Korea, N=387	USA, N=440	China, N=421
Gender	Male	47.18%	58.75%	44.09%	39.43%
	Female	52.82%	41.25%	55.91%	60.57%
International interactions	Yes	67.41%	32.75%	65.00%	70.07%
	No	32.59%	67.25%	35.00%	29.93%
		Mean (SD)	**Mean (SD)**	**Mean (SD)**	**Mean (SD)**
Age (years)		39.70 (11.36)	44.08 (13.52)	41.48 (9.89)	33.43 (6.35)

APPENDIX C

Table C1 reports the weights and significance levels of the formative indicators for each construct across the three countries: South Korea, the USA, and China. Most indicators are statistically significant, indicating meaningful contributions to their respective constructs. Therefore, all indicators were retained to preserve the conceptual coverage of the constructs, consistent with prior literature (Luo et al., 2020; Hair et al., 2022).

Table C1. Weights and p-values for Constructs

Constructs	South Korea	USA	China
	Weight (p-value)	Weight (p-value)	Weight (p-value)
Properties of email ID			
Communicability (V1)	0.309 (0.265)	0.171^{*} **(0.014)**	0.028 (0.858)
Global Reach (V2)	0.169 (0.543)	0.321^{***} (**0.000**)	0.289^{*} (**0.040**)
Memorability (V3)	0.140 (0.524)	0.286^{***} (**0.000**)	0.446^{**} (**0.006**)
Anonymity (V4)	0.162 (0.270)	0.434^{***} (**0.000**)	0.012 (0.924)
Recognizability (V5)	0.719^{***} (**0.000**)	0.402^{***} (**0.000**)	0.616^{***} (**0.000**)
Properties of bMail ID			
Identifiability (V6)	0.204^{*} (**0.021**)	0.495^{***} (**0.000**)	0.393^{***} (**0.000**)
Verifiability (V7)	0.394^{***} (**0.000**)	0.303^{**} (**0.001**)	0.472^{***} (**0.000**)
Sustainability (V8)	0.625^{***} (**0.000**)	0.407^{***} (**0.000**)	0.496^{***} (**0.000**)
Gain Trust as Sender			
MD1	0.534^{***} (**0.000**)	0.469^{***} (**0.000**)	0.437^{***} (**0.000**)
MD2	0.503^{***} (**0.000**)	0.595^{***} (**0.000**)	0.690^{***} (**0.000**)
Social Properties of email ID			

Communicability with Senders (VS1)	0.318 (0.256)	0.395*** (**0.000**)	0.520*** (**0.000**)
Globally Visible to Senders (VS2)	-0.654* (**0.018**)	0.299* (**0.017**)	0.342* (**0.036**)
Being Remembered by Senders (VS3)	0.851** (**0.003**)	0.265* (**0.032**)	0.171 (0.261)
Recognizable Senders (VS5)	0.603*** (**0.000**)	0.312*** (**0.000**)	0.354** (**0.008**)
Social Properties of bMail ID			
Identifiable Senders (VS6)	0.626*** (**0.000**)	0.399*** (**0.000**)	0.427*** (**0.000**)
Verifiable Senders (VS7)	0.510*** (**0.000**)	0.422*** (**0.000**)	0.444*** (**0.000**)
Sustainable Senders (VS8)	0.037 (0.570)	0.374*** (**0.000**)	0.483*** (**0.000**)
Build Social Trust as Receiver			
MDS1	0.555*** (**0.000**)	0.416*** (**0.000**)	0.469*** (**0.000**)
MDS2	0.485*** (**0.000**)	0.628*** (**0.000**)	0.675*** (**0.000**)

Note: $^{*}p < 0.05$, $^{**}\ p < 0.01$, $^{***}\ p < 0.001$.

We adopted two approaches to assess and mitigate potential Common Method Variance (CMV) bias.

(1) We employed the full collinearity VIF test (Kock, 2015). The results in Table C2 demonstrate high robustness across datasets: most VIF values for the constructs in the China and South Korea samples were below the conservative threshold of 3.3. In the USA data, while the VIF values for two constructs (3.498 and 3.421) slightly exceeded 3.3, they remained well below the more lenient acceptable limit of 5.0 (Hair et al., 2022).

Table C2. Full Collinearity VIF Assessment

Constructs	VIF		
	South Korea	USA	China
Properties of eMail ID	1.177	1.099	1.016
Properties of bMail ID	1.048	1.253	1.101
Resilience of Platform	1.359	1.993	1.329
Gain Individual Trust as Sender	1.207	1.499	1.348
Social Properties of eMail ID	1.192	1.194	1.156
Social Properties of bMail ID	1.049	1.520	1.697
Build Social Trust as Receiver	1.135	2.011	1.202
Sender's Adoption of bMail ID	2.013	2.922	1.448
Receiver's Adoption of bMail ID	3.001	3.498	1.735
User Adoption of bMail ID	2.880	3.421	1.562

(2) We applied the marker variable approach (Malhotra et al., 2006). By partialing out the second-smallest positive correlation (r_M) from the original correlation matrix, we found that all structural paths remained significant after adjustment (results are shown in Appendix Table C3).

Table C3. Marker Variable Results

Path	South Korea		USA		China	
	Base Model	Marker Model (r_M=0.009)	Base Model	Marker Model r_M=0.148	Base Model	Marker Model r_M=0.015
Properties of eMail ID → Sender's Adoption of bMail ID	0.263***	0.256***	0.284***	0.160***	0.178**	0.165***
Properties of bMail ID → Gain Individual Trust as Sender	0.594***	0.590***	0.719***	0.670***	0.547***	0.540***
Resilience of Platform → Sender's Adoption of bMail ID	0.160**	0.152**	0.245***	0.114*	0.234***	0.222***
Gain Trust as Sender → Sender's Adoption of bMail ID	0.321***	0.315***	0.280***	0.155**	0.181**	0.169***
Social Properties of eMail ID → Receiver's Adoption of bMail ID	0.187***	0.180***	0.284***	0.160***	0.291***	0.280***
Social Properties of bMail ID → Build Social Trust as Receiver	0.749***	0.747***	0.728***	0.681***	0.519***	0.512***
Build Social Trust as Receiver → Receiver's Adoption of bMail ID	0.506***	0.502***	0.439***	0.342***	0.302***	0.291***
Sender's Adoption of bMail ID → User Adoption of bMail ID	0.262***	0.255***	0.312***	0.192***	0.243***	0.231***
Receiver's Adoption of bMail ID → User Adoption of bMail ID	0.598***	0.594***	0.568***	0.493***	0.374***	0.364***

Note: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

We conducted the Measurement Invariance of Composite Models (MICOM) procedure across the three countries (Henseler et al., 2016). The results, based on the correlation and 5% quantile, support conducting a multi-group analysis, as shown in Table C4. Otherwise, it is denoted as "not comparable" in Table 6.

Table C4. MICOM Results

Construct	Correlation			5% Quantile		
	S. Korea vs. China	S. Korea vs. USA	USA vs. China	S. Korea vs. China	USA vs. S. Korea	USA vs. China
Properties of eMail	0.953	0.878	0.873	0.835	0.898	0.923
Properties of bMail	0.981	0.958	0.990	0.957	0.969	0.967
Gain Individual Trust as Sender	0.995	0.999	0.999	0.963	0.982	0.979
Social Properties of eMail ID	0.711	0.778	0.996	0.705	0.827	0.929
Social Properties of bMail ID	0.934	0.961	0.999	0.963	0.977	0.968
Build Social trust as Receiver	0.995	0.997	1.000	0.979	0.990	0.987

Note: A correlation larger than the 5% quantile indicates that compositional invariance is established.